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Egyptian Verbal Classes

A Shift in Paradigm

Marwan Kilani[1]

Abstract

This article critically re-evaluates the traditional classification of Egyptian verbal roots, a foundational concept in Egyptian and Coptic linguistics that has remained largely unchallenged since its introduction over a century ago. Although widely used in grammars, dictionaries, and historical linguistic studies, no systematic reassessment of their definition has ever been undertaken since their adoption. This study addresses this gap by examining the structure of roots, stems, and morphological patterns across Egyptian and Coptic. It introduces a new notational system designed to represent the verbal system more accurately and argues for a redefinition of verbal classes based on greater internal coherence. The analysis demonstrates that several established categories, including III-weak, II-gem, and IV-weak, are not independent classes but rather paradigmatic variants of strong bi- (or 2 strong + 1 weak) and tri-consonantal roots. The revised model provides a simpler and more consistent framework, with broad implications for the study of Egyptian morphology, phonology, and historical linguistics.

1 Introduction

Egyptian verbal roots are traditionally grouped into verbal classes based on the number and types of consonants that are assumed to constitute them. These verbal classes lie at the very core of our modern understanding of the Egyptian and Coptic languages: they are employed in every contemporary grammar and dictionary, are central to the description and analysis of Egyptian morphology, and form a crucial framework in current linguistic studies on phonology, vocalization, and Egyptian historical linguistics more broadly. Yet, as surprising as it may seem, verbal classes and the validity of their definitions have never been systematically reassessed by modern Egyptologists since they were first introduced over a century ago,[2] and they became an unchallenged assumption that has shaped, and perhaps hindered, our understanding of the Egyptian verbal system up to the present day.

The present article seeks to challenge this state of affairs. Beginning with a systematic reassessment of roots, stems, and morphological oppositions as they can be observed in Egyptian and Coptic, this study aims first to propose a new notational framework

1 University of Basel (kilani.edu[at]gmail.com). Part of the SNSF Grant PZ00P1_201964.

2 The earliest use of verbal classes I am aware of is in Erman's grammar (1894) and the earliest detailed discussion in Sethe's book on Egyptian verbs (1899).

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for accurately representing the Egyptian verbal system, and second to redefine verbal classes in a more coherent and linguistically robust manner. As will be shown, the model presented here differs significantly from the traditional one, while offering greater internal consistency and simplicity. In particular, it will be argued here that the traditional pre-Coptic verbal classes known as III-weak, II-gem, and IV-weak in fact do not exist as independent categories, and should instead be analysed as mere paradigmatic variants of strong two- (or two strong + one weak - see below) and three-radical roots.

Given the central role of verbal classes in Egyptian linguistics, this revised model has wide-ranging implications and opens several promising directions for future research.

2 Structure of the Paper

The paper is structured as follows. First, a series of conventions for the transcription of Egyptian, Coptic, and reconstructed forms is introduced. This is followed by a brief discussion of the current assumptions regarding verbal classes, along with an outline of the problems that characterize them. The article then turns to the definition of a new theoretical and terminological framework. A revised model for accurately describing roots and stems, both synchronically and diachronically, is first presented. This is followed by an analysis of non-root and non-stem morphological elements and oppositions, with particular focus on the infinitive forms preserved in Coptic. Other verbal forms remain beyond the scope of this study and will need to be addressed in future work. This section culminates in a complete analytical framework for Egyptian/Coptic infinitives, as well as a set of observations regarding the number and types of verbal classes that must have existed in pre-Coptic Egyptian.

The final section summarizes the evidence presented in the previous parts of the article, arguing that it provides strong evidence for the existence of only two main verbal classes for verbs with less than four strong consonants: one characterized by two strong consonants and possibly a weak one, and another by three. Additional classes involving more strong consonants may have existed, but are difficult to reconstruct with confidence due to the scarcity of available evidence. The article concludes with a few final remarks and a full paradigm of the infinitives according to the new model, including Coptic examples for each form.

3 Conventions and Abbreviations

The following notations are used in the article:

In the transliteration of pre-Coptic Egyptian:

- I use an Egyptological transliteration based on Gardiner's, with *i̯* as a conventional sign for the consonant corresponding to  and *q* for for  .
- { } are used for consonants that are attested in hieroglyphic spellings but had likely already been lost in the contemporary pronunciation of the words.

In the transcription of Coptic:

Dialects are indicated within square brackets [ ] ; the abbreviations used are the same as those employed by Crum in his dictionary (Crum 1939).

- I use the notation | | to signal quasi-phonological *interpretations* of Coptic forms. I describe these as "interpretations" rather than "transcriptions" or "reconstructions" for the following reasons: they are not reconstructions, which would be marked with *, since they correspond to attested forms rather than hypothetical ones; they are also not full phonological transcriptions, which would be marked with / / or [ ], because the phonology of historical Coptic, while generally understood, is less securely established than that of a living language and continues to be debated in certain respects. The forms presented in | | therefore reflect *my interpretations* of the likely pronunciation of the corresponding Coptic forms (based mainly on Loprieno 1995; Peust 1999). It should be emphasized, however, that these *interpretations* are not definitive, and other scholars may disagree with particular details (on these issues, see Müller 2011).
- Besides the obvious equivalences, the following conventions need mention:
  ⲱ = |o|, ⲟ = |ɔ|, ⲏ = |e|, ⲉ in stressed position = |ɛ|, ⲉ in unstressed position = |ə|, ⲓ as vowel in stressed position = |i|, ⲓ as vowel in unstressed position = |i̯|,[3] ⲓ as consonant = |j|, ⲟⲩ as vowel = |u|, ⲟⲩ as consonant = |w|, ⳕ = |kʲ|, ⲭ = |č|, ϩ = |h|, ϧ = |h₂|
- Stress is marked with ´
- Vocalic/Syllabic consents are marked with ̩
- Vowels written twice are interpreted as long, and indicated with : e.g. ⲁⲁ = |a:|. Note that other scholars interpret double vowels as indicating a glottal stop, e.g. ⲁⲁ = |aʔ|. The use of | | signals explicitly that these are interpretations subject to debate, not established phonological facts.
- The sign ᵊ indicates the potential presence of an unstressed vowel in a position where such vowel is not explicitly indicated in the corresponding Coptic spelling.

In reconstructions:

- C = unspecified/any consonant
- V = unspecified/any vowel
- The position of a consonant in a root is indicated with a superscript number; e.g. C^1 = 'first radical consonant'. This can be used to indicate reduplication/gemination; e.g. $C^1C^2C^3C^2C^3$ = 'form from a three-consonant root with reduplication of the last two consonants' or also $C^1C^2C^2$ = 'form from a two-consonant root with reduplication/gemination of the last consonant'
- *y̯* = unspecified semiconsonant/glide = *j̯, w̯, ʔ*
- ***a*** = ancestor of Coptic ⲱ = |o|, ⲟ = |ɔ|, and ⲟⲩ = |u| (and allophonic and dialectal variants)

3 The use of i̯ here is partially conventional, and it is meant to represent a unstressed and likely reduced (mostly final) vowel in those words and dialects where it is spelled with ⲓ rather than the more common ⲉ.

- **E* = ancestor of all the other Coptic vowels (see below for discussion)
- I mark with □ forms that can be expected to have existed but which are not currently attested, such as e.g. the expected absolute form of a Coptic infinitive which is attested only in its prenominal and prepronominal forms.

In glossae:

- ⟨ ⟩ are used to gloss a feature encoded by the vowel in the frame of non-concatenative morphology; e.g. the reconstructed form **p⟨É⟩r* is glossed as 'moving_forth.⟨NONACT⟩', were the consonants *pr* encode 'moving_forth' while the vowel *É* encodes the feature 'NONACT'.
- NONACT = non-actant experiencer

Note: Additional abbreviations and conventions are introduced throughout the article as part of the new model presented here. Each is discussed where it first appears.

4 Starting from the Bases: Deconstructing Verbal Classes

Nearly all modern grammars of pre-Coptic Egyptian categorize verbs into classes such as 2-literal, 3-literal, III-weak (or III-infirmae), and so on. At first glance, the meaning of these classes may seem obvious, and as a result, many modern Egyptological grammars provide little explanation of what they actually represent, and usually do not discuss the origins of this classification or the reasonings and assumptions underlying their original definition. For instance, Allen's grammar simply states: "Egyptologists divide Egyptian verbs into classes based on the form of their root," without any further elaboration on their scholarly history or on the conceptual assumptions underlying them (Allen 2014: 167).

Allen's description reflects how verbal classes are generally understood in contemporary Egyptology: they are seen as a way to categorize verbs based on their consonantal roots, in a manner largely analogous to the verbal classes in Semitic languages. For example, a verb like *mn* is classified as 2-lit because its root is assumed to have only two strong consonants. At the same time, verbs classified as 2-lit are assumed to have had only 2 strong consonants in their root because they are classified as 2-lit—essentially creating a circular logic where the classification and the assumption reinforce each other without deeper explanation.

This is a crucial issue because verbal classes form an implicit yet fundamental framework for all modern analysis of the Egyptian verbal system and Egyptian as a whole, including the reconstruction of its vocalic patterns. Scholars not only organize their reconstructions according to these classes but also shape them to fit the root forms suggested by the classes themselves. This approach presupposes that Egyptian verbal classes, as defined today, accurately represent the original structure of verbal roots. However, this assumption can be easily challenged, and in fact, it was not the original intent of verbal classes when they were first introduced.

So, what exactly are verbal classes, and what were they intended to represent? Answers to these questions can be found in the earliest publications that discuss them in detail,

particularly Sethe's book on Egyptian verbs (Sethe 1899) and Gardiner's Grammar of Middle Egyptian (Gardiner 1957 and already in the earlier editions), which is largely based on the former, though not always explicitly. I cite Gardiner's grammar instead of Sethe's study because Gardiner's text is more concise and better structured, effectively illustrating the main issues at hand. Gardiner states:

A. "A classification of the verbs is therefore required, and the basis of this must be the mutability or immutability of the stem. A second mark serving to distinguish the different verbal classes is the gender of the infinitives, some classes having masculine infinitives like  *ʿnḫ* 'to live', while others have infinitives showing the feminine ending *-t*, ex.  *mst* 'to bear', 'to give birth'." (Gardiner 1957: 206)

But then:

B. "Classification according to number and nature of the radical consonants – We shall now proceed to classify the different kinds of Egyptian verb-stems, premising that only such distinctions will be noted as may prove useful in the study of Egyptian texts. Coptic shows that adjective-verbs like *sbŏk* 'to be small' were vocalized otherwise than transitive verbs like *sōtᵊm* 'to hear', but such facts as these must be ignored in this grammar, since they cannot be followed up in the hieroglyphs." (Gardiner 1957: 212)

Moreover, specifically in respect to 2-lit verbs:

C. "As in the Semitic languages, the typical verb-stem consists of three radical consonants, ex.  *sḏm* 'hear'. There existed, however, a considerable class of biliteral stems, ex.  *mn* 'remain', 'endure', though it may be shown that many of these originally belonged to one or other of the classes with three consonants, such as  *wḏ* 'command', Arabic وصى (*waṣa*) ;  *tm* 'be closes', Arabic تمّ (*tamma*). Whereas the verbs just quoted had, owing to some inherent weakness, passed in prehistoric times from triconsonantal to the biconsonantal class, there are other apparently biliteral stems, such as  *m(w)t* 'die', which prove on closer inspection to be really triliterals." (Gardiner 1957: 210)

D. "Biliteral verbs, abbreviated 2-lit., exx.  *wn* 'open',  *mn* 'be firm'. The infinitives are masculine, and of the form quoted. For the originally triliteral character of these verbs see above § 273 ; and that many of them may be derived from tertiae infirmae (below §281) is perhaps hinted by the feminine infinitives of their causatives (above §275 and below §282)." (Gardiner 1957: 213)

And more in general:

E. "The basis of our classification will be the number of radical consonants, whether weak or strong, single or reduplicated. The designations of these classes are those usually adopted, though they are not altogether satisfactory. By *biliteral, triliteral, quadrilateral*, and *quinqueliteral* verbs are meant those having two, three, four, or five immutable (strong) consonants respectively, though there is really no reason (e.g.) for refusing the name 'triliteral' to triconsonantal stems with identical second and third radicals (the *secundae geminatae* class, §280) or to those having a weak third radical

(the *tertiae infirmae*, §281). The notion of gemination inherent in the names *secundae* and *tertiae (litterae) geminatae* is also misleading, since both here and in the *tertiae* and *quartae infirmae*, so far from gemination or doubling being employed for making twofold a consonant that was originally single, its presence actually warns us that the verb-stem in question possessed from the beginning a final radical letter which was specifically prone, either from inherent weakness or from its identity with the penultimate, to disappear from writing." (Gardiner 1957: 212-213)

As for so-called weak or *tertiae infirmae*, he states:

F. "In the case of the weak verbs ending in  *i* or  *w* matters are complicated by the fact that these consonants (or semi-vowels, §20) were often omitted in writing." (Gardiner 1957: 208)

It is clear that the picture that emerges from these passages is significantly more complex and, indeed, more confusing than what is suggested by the one-sentence introduction to verbal classes found in Allen's grammar. Several points and issues deserve to be highlighted:

1) As indicated in passage A, verbal classes were intended to be defined based on *formal* alternations among different *morphological* contexts. In other words, these classes were originally conceived as *grammatical paradigms* rather than as a classification of the underlying consonantal roots.
2) Yet, as passages B and E indicate, the actual classification of verbs into these paradigms is based primarily on the number and types of consonants of the root of the verb. This choice is puzzling, and results in classifications that are explicitly going against the goals stated in passage A. For instance, Gardiner states that the supposed weak consonants of III-inf verbs are often omitted (passage F), but this does not seem to prevent him or later scholar to make a class characterized by such often purely hypothetical root consonants out of these verbs.
3) At the same time, the criteria determining the "number and nature of the radical consonants" (passage B) used in the classification of these verbs is also confused and incoherent, certainly not comparable to the way root consonants are defined and used in Semitic, where root consonants constitute an abstract underlying structure, independent of how particular verbal forms are represented in texts. For instance, passages C and D clearly show that verbs are classified as 2-literal whenever their Middle Egyptian *spelling* displays only two strong consonants, even though comparative Afro-Asiatic (e.g. Semitic) or Coptic evidence (or both) might suggest the presence of a third weak consonant in the original root. Sethe is even more explicit in this case, as for several verbs that he classifies as 2-literal, he explicitly indicates in parentheses where a third unattested weak consonant might have been located within the root (see e.g. Sethe 1899: 224).

   This clearly shows that the "radical consonants" used in the classification of these verbs should not be understood as the consonants that were likely part of the original, proto-Egyptian-Coptic root, nor as the consonants—written and unwritten—that were likely

part of the root in Middle Egyptian. Instead, they represent only the consonants that appear *in writing* in the (mostly Middle Egyptian) spellings of the word...

4) …except when they do not, as in the case of III-inf verbs. Clearly, verbs can be classified as III-inf, even though the final weak consonant that should characterize them is never spelled out in the attested forms, as appears from passage F.

Therefore, far from being a rational and systematic way of classifying Egyptian verbs, the traditional system of verbal classes appears as a rather chaotic conflation of at least three different linguistic realities, namely:

– paradigmatic alternations across different morphological forms
– number and types of root consonants as appearing in writing at specific points in time
– inclusion of certain hypothetical root consonants that may not be attested in Middle Egyptian, while other consonants, unattested in Middle Egyptian but present in Coptic or other Afro-Asiatic languages, are excluded.

Due to these issues, the resulting system (or lack thereof) appears fundamentally flawed on two distinct levels. First, at a purely terminological level, it fails to accurately and unambiguously describe the complex interaction between roots and their surface realizations, conflating the two irreparably. Second, the traditional system is also likely to be linguistically inaccurate: if the classes it is based on are defined on questionable premises, they are unlikely to represent real underlying linguistic realities. This contribution seeks to address both issues: first, by proposing a new notational framework to accurately represent the Egyptian verbal system, and second, by redefining verbal classes in a more coherent and linguistically robust manner.

5 Beyond Traditional Verbal Classes – A New Terminological Framework

It has been shown in the previous section that the traditional verbal classes are ambiguous and rife with uncertainties, incongruences, and *ad hoc* solutions, making them an inadequate framework for analysing verbs or reconstructing their vocalization. This section proposes a new terminological framework intended to better reflect the specific features of the Egyptian verbal system as objectively as possible. The model developed here is organized into three main components. The first is a new conceptualization of what constitutes a "root", along with a set of terms to represent roots accurately in both synchronic and diachronic perspectives. The second component is a discussion of what I refer to as semantic modifiers of the root. These are modifications that affect the meaning of the root in a more fundamental way. They include semantic reduplication and stem derivation. The third component concerns morphological modifiers that operate at a more superficial level. These elements influence how specific semantic features of a root or stem (e.g. diathesis or the like), are expressed. This group includes the vocalic opposition **a/*E*, the vocalic opposition −V̄ / +V̄, the suffix *-ỳ*, and the suffix *-(V)t* used in infinitives.

5.1. Roots

5.1.1. Etymological Root vs Synchronic Root

Roots and root consonants are a theoretical construct, as roots in their abstract form are never directly attested. However, they are also "real" in the sense that they provide both the semantic base and the consonantal skeleton of actual, attested verbal forms. Root consonants are the consonants associated with the fundamental meaning of a root.

Roots are both a synchronic and a diachronic phenomenon, and one of the shortcomings of the traditional verbal class system is its lack of temporal sensitivity, namely it does not attempt to relate the classes it assumes for specific verbs to specific historical phases of the language. As a result, some definitions reflect a synchronically specific reality, while others are based on an absolute or "original", etymological interpretation of the verb's form. For instance, as seen above, so called 2-lit verbs are classified as such because in some of their attested forms they display only 2 consonants, even though there could be reasons to assume that *originally* or *etymologically* other consonants might have been present; by contrast so-called III-inf are assumed to have had an *original* or *etymologically* weak consonant, even though for many verbs grouped into this class such consonant is never consistently written and may in fact have been lost (if it were there at all to begin with) in the attested phases of the language.

I suggest to solve this ambiguity by introducing a distinction between:

A) ETYMOLOGICAL ROOT / ETYMOLOGICAL ROOT CONSONANTS
B) SYNCHRONIC ROOT / SYNCHRONIC ROOT CONSONANTS

ETYMOLOGICAL ROOT CONSONANTS correspond to the consonants that can be reconstructed as the diachronic source and proto-root of a set of cognates attested in one or more specific phases of the language. Etymological roots can be organized into etymological root classes on the basis of the number of their consonants. I suggest to abbreviate etymological root consonants classes as follows: 1-EtR = 'etymological root with 1 consonant', 2-EtR = 'etymological root with 2 consonants', 3-EtR = 'etymological root with 3 consonants', etc., and I suggest to use the sign √ to indicate etymological roots in opposition to their synchronic descendants or conjugated manifestations – for instance √*wꜣḏ* : '(to) be(come) fresh/green' = [etymological root] vs *wꜣḏ* = 'to be(come) fresh/green' [conjugated form], e.g. infinitive. Note that I propose marking the distinction between the root and the conjugated form in translation as well. Specifically, I suggest translating roots and other non-conjugated forms into as English infinitives with 'to' in parentheses, while using the full infinitive with 'to' only for the actual corresponding infinitive.

In some cases, all the etymological root consonants and their order can be reconstructed with reasonable certainty either internally or through the comparison with other Afro-Asiatic languages. For instance, the etymological root √*wꜣḏ* is likely to be a 3-EtR root already at the proto-Egyptian level not only because of the Egyptian evidence, but also because of its clear relation with the Semitic root √*wrq* (see e.g. Vycichl 1984, 238; Loprieno 1995, 31–32); we have no reason here to postulate any additional unattested root consonant.

In other cases, however, we may be certain only about *some* root consonants, while the existence of others remains hypothetical. For instance, the Egyptian forms of the verb  *wḏ* (inf) ‘to command’ suggest it had only two root consonants. However, if the scholars who connect this Egyptian verbal with Arabic وصى = √*wṣj* : ‘(to) ask’, ‘(to) command’ and see in the Arabic form a 3-rad root are right (e.g. Gardiner 1957: 210),[4] then it is possible that the original form of the Egyptian verb included an additional consonant that corresponded to the Arabic *-j*, but which was entirely lost by the time this verb was attested in Egyptian. At the same time, the existence of this consonant remains purely hypothetical, and reconstructing it in Egyptian would be unjustified, as in principle it is also possible that the Egyptian form, with only two consonants, represents the original state, while the additional *-j* in Arabic could have been a later innovation specific to Semitic. In these cases, I propose using the notations 2^{+}-EtR, 3^{+}-EtR, etc., to indicate that the root in question may have included one or more additional consonants (hence the $^{+}$) beyond the one, two, three, etc., that are attested. Similarly, the $^{+}$ can also be used in the notation with √. For instance, √*pr*$^{+}$ would denote a verb which had *p* and *r* as attested etymological root consonants, and which might have had additional, unspecified consonants. If there is some indication of the form of the hypothetical additional consonant but not its position within the root, this consonant can be indicated after the $^{+}$. For example, √*pr*$^{+}$*w* would represent a root with *p* and *r*, and possibly *w*, though the original placement of the latter within the root is uncertain (e.g., it could be √*wpr*, √*pwr*, or √*prw*).

Finally, etymological root consonants by definition are assumed to predate the forms in which they are attested. Therefore, the earliest attested forms serve as a *terminus ante quem* for their existence, situating these roots in a pre-X phase of the language. For example, let us imagine: a root reconstructed solely from Coptic evidence (whether it is because of the absence of attestations in earlier stages or because of other reasons) would be by definition a proto-Coptic root, and would have to be conceptually placed in a generic pre-Coptic phase of the language. Conversely, a root attested for example by both Coptic and Middle Egyptian evidence would have to be understood as more specifically proto-Middle-Egyptian, and could be situated in a more specific pre-Middle Egyptian phase, as the Middle Egyptian evidence would be the earliest evidence used. However, this would not imply that roots labelled as e.g. proto-Middle Egyptian are necessarily older than those labelled as e.g. proto-Coptic: in fact, since both categories have chronological open lower boundaries, and their earliest attestations set only an upper limit, in principle their actual ages could overlap and extend far earlier than the surviving evidence suggests.

Since etymological roots as defined here are not absolute, but have specific chronological boundaries, it is crucial to explicitly mark the chronological context of an etymological root. In order to do so, I suggest to append a superscript temporal label to the √-form or to the class abbreviation. For instance, the Old Egyptian verb *wḏ* (inf) : ‘(to) be(come) fresh/green’, derives from a 3-EtR^{prtOE} root (^{prtOE} = ‘proto-Old-Egyptian’) having the form

4 Note that I do not mean to endorse this comparison, which would require a detailed discussion beyond the scope of this paper; I cite it here merely as an illustrative example drawn from previous scholarship.

√*wꜣḏ*^{prtOE}. Codes like the following ones can be used to situate etymological roots in time: $^{\text{pr-prtEg}}$ = pre-proto-Egyptian (as a generic prehistoric ancestor of the earliest proto-phase that can be reconstructed on the Egyptian evidence alone), ^{prtCpt} = proto-Coptic, $^{\text{pr-prtCpt}}$ = pre-proto-Coptic, ^{prtOE} = proto-Old-Egyptian, ^{prtME} = proto-Middle-Egyptian, etc.

Moreover, on the basis of comparisons with other Afro-Asiatic languages, it has been suggested that Egyptian *ḏ* < ***q* and *ꜣ* < ***r* in prehistoric times, and it would be good to include these data in our analysis (I explain the ** here below). However, here another notational distinction is needed, as there is a conceptual difference between *ḏ* and *ꜣ*, which are transcriptions of attested phonemes according to specific Egyptological conventions, and ***q* and **r* which are hypothetical, reconstructed pre-Egyptian phonemes that might well fall outside the range of phonemes covered by the Egyptological transliterations. This latter point obviously touches on the thorny question that Egyptological transliteration conventions may be efficient for transcribing ancient Egyptian but are often very idiosyncratic and hard to reconcile with non-Egyptological practices. I suggest two additional conventions to address these issues. The first is to add a double ** after the √ to indicate that the corresponding root is a reconstructed hypothesis that takes into consideration other extra-Egyptian Afro-Asiatic languages, and as such it might include phonemes that may not be directly attested in *any* Egyptian phase of the language. The choice of marking it with a double ** instead of a single * derive from the idea that a single * is usually adopted to indicate protoforms, but if we define as proto-Egyptian the forms that can be reconstructed solely on the basis of internal Egyptian evidence (see e.g. discussion of the internal reconstruction of vowels here below), then the hypothetical forms discussed here must be *pre*-proto-Egyptian, as they are reconstructed on the basis of external evidence that allows us to reach a historically deeper level than what would be possible on the basis of the Egyptian evidence alone; hence the use of the double **. The second convention is to use (and to assume the use of) more generic non-Egyptology-specific notations (ideally IPA-based, or at least aligning with the most common practices in Afro-Asiatic linguistics when distinct form IPA standards) for the transcription of these reconstructed protoforms. Since these protoforms are grounded in Egyptian data yet transcribed with non-Egyptological conventions, they can serve as bridges and interfaces between Egyptology and Afro-Asiatic linguistics. All this considered, we can start from the form √*wꜣḏ*^{prtOE} and by applying the changes *ḏ* < ***q* and *ꜣ* < ***r* we can also reconstruct a pre-proto-Egyptian root √***wrq*$^{\text{pr-prtEg}}$. When an etymological root is assumed to derive from an another, earlier one, the sign > can be used: √***wrq*$^{\text{pr-prtEg}}$ >√*wꜣḏ*^{prtOE}.

By contrast, I define Synchronic root consonants as the consonants corresponding to the consonantal skeleton of a verb at a specific point in time, in a specific *attested* phase of the language. Verbs can also be grouped into classes on the basis of their synchronic root consonants; these classes will be abbreviated here as 2-SyR, 3-SyR etc. and they will be introduced by the sign Ⳇ , e.g. Demotic Ⳇ*wt* .

Synchronic root consonants represent the morphological root of the verb within the respective synchronic phase of the language – i.e. they are the root that are used in a specific phase of the language to construct contemporary productive verbal forms. These consonants may originate from earlier etymological root consonants or from the combination of

etymological roots with morphological affixes or other kinds of morphological modifiers (e.g. consonant reduplication) that were productive in earlier stages of the language but are no longer so in the synchronic phase. However, they do not include form-specific morphemes, such as the personal endings of the pseudoparticiple/stative or the *-t* ending of feminine participles. A good illustrative example is provided by the causative prefix *s*-, which might have still been productive in Old Egyptian, but not anymore in Demotic. As a result, the Old Egyptian synchronic root of the Old Egyptian verb *sʿnḫ* (inf) : 'to cause to live' can be defined as ⌞*ʿnḫ*, with an extra-root prefix *s*-, while the demotic synchronic root of the Demotic verb *sʿnḫ* (inf) : 'to cause to live' can be defined as ⌞*sʿnḫ*, as by that time the causative prefix was not productive anymore, and synchronically *sʿnḫ* behaved as a 4-consonant verb in that phase of the language. Naturally, all these forms would ultimately derive from a proto-Old-Egyptian etymological root √*ʿnḫ* which might in turn derive from an earlier pre-proto-Egyptian form √***dnḫ*, if the scholars who suggest a ***d* > *ʿ* development in the prehistory of Egyptian are correct.

Synchronic roots may also differ from the etymological roots due to the absence of specific consonants that were lost at some point in the language's history. For instance, the proto-Old-Egyptian etymological root √*bꜣk* (3-EtR) : '(to) work' is synchronically realized as ⌞*bꜣk* (3-SyR) in Old and Middle Egyptian, but as ⌞*bk* (2-SyR) in Late Egyptian and Demotic, as *ꜣ* is regularly lost in that position in later phases of the language.

Finally, consonants in synchronic roots can differ from those of the corresponding etymological root in both form and position due to historical phonological changes. For example, the synchronic root of the Coptic verb [S] ⲥⲟⲟⲩⲧⲛ |sṓwtˢn| (infinitive) : 'to stretch' can be described as ⌞*swtn* (4-SyR), which ultimately derives from a pre-proto-Egyptian etymological root √***ṯwn/l*[5] : '(to) be(come) long' through the following developments within Egyptian:

a) lexicalization of the causative prefix *s*-,
b) metathesis ***ṯw* > ⲟⲩⲧ |wt|

Naturally, several distinct synchronic roots can emerge out of a same single etymological root. For instance, the following synchronic roots are all derived from the pre-proto-Egyptian 3-EtR root √***wrq*:

√***wrq* = 3-EtR$^{\text{pr-prtEg}}$
>√*wꜣḏ* = 3-EtR^{prtEg}
⌞*wꜣḏ/t*[6] : '(to) be(come) fresh/green' = 3-SyR (Middle Egyptian)
⌞*swt* : '(to) make fresh/green' = 3-SyR (Demotic)

As in the case of etymological roots, it is crucial to explicitly specify the particular phase of the language to which a synchronic root belongs.

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5 With *ṯ* as a non-Egyptology-specific transcription of the pre-proto-Egyptian ancestor of the Egyptian consonant transcribed as *d* in the Egyptological transliteration system adopted here (as external, non-Egyptian evidence for this reconstruction, cf. Arabic *ṭyl* = 'to be long').

6 There is evidence that *ḏ* had developed into *t* already in Middle Egyptian, although this development may have been limited to some lexical items and/or dialects (see Kilani 2021).

Here as well, I propose to do that by appending indexes to the synchronic root label: e.g. 3-SyR^{Cpt} indicates a Coptic synchronic root that is characterized by 3-consonant. By contrast, 2-SyR^{LE} would indicate a synchronic root with 2 consonants in Late Egyptian. Chronological labels can also be prefixed to the sign ḫ , to provide the chronological context of a specific synchronic root; e.g. ^{ME}ḫ*bꜣk*: '(to) work' vs ^{LE}ḫ*bk* : '(to) work'. The following labels can be used to refer to the main attested phases of the language: ^{Cpt} = Coptic, ^{OE} = Old Egyptian, ^{ME} = Middle Egyptian, ^{LE} = Late Egyptian and so on.

This system not only provides an accurate and unambiguous way to represent synchronic roots in all stages of the language, but it also allows the accurate and unambiguous notation of both attested and reconstructed diachronic changes in diachronically related synchronic roots. In this case, when several synchronic roots form a diachronic sequence, I will use ḫ to indicate the earliest instance of the root, and >ḫ for its later iterations. For instance:

√***wrq* = 3-EtR$^{pr\text{-}prtEg}$
>√*wꜣḏ* = 3-EtR^{prtEg}
ḫ*wꜣḏ* = 3-SyR^{OE} : '(to) be(come) fresh/green'
>ḫ*w{ꜣ}ḏ/t* = 2-SyR^{LE} : '(to) be(come) fresh/green'
>ḫ*wt* = 2-SyR^{Cpt} : '(to) be(come) fresh/green'

5.2 Root Reduplication

5.2.1 Reduplication

Egyptian verbal forms can be divided into two categories based on whether they involve partial or full reduplication of the root consonants. Egyptologists typically classify reduplication into three typological categories (see Bendjaballah and Reintges 2009 for an overview): 1) Lexical reduplication, which is thought to transform a basic verb into its pluractional counterpart; 2) Morphological reduplication, which is believed to express certain TAM features (especially imperfectiveness) in specific verbal forms and traditionally defined verbal classes; 3) Structural reduplication, which is supposed to be a structural feature of some verbal roots without any specific meaning. Only the first type, lexical reduplication, is intended here; certain types of morphological and structural reduplication in infinitives will be discussed later in the article.

Lexical reduplication is a relatively straightforward feature that manifests in a binary way: roots are either reduplicated or not. Based on both Egyptian and Coptic evidence, lexical reduplication typically involves only two consonants of the root. When the root appears to have only two synchronic consonants, both consonants are reduplicated. When the root has more than two synchronic consonants, only the last two consonants are reduplicated.[7] As for notation, I propose marking reduplication with a superscript $^{(+R)}$ when present, and $^{(-R)}$ when absent.

7 I consider the occasional cases in which roots with three consonants appear to be fully reduplicated better explained as instances of word repetition rather than true lexical reduplication, as these occurrences are rare, and Coptic provides no evidence that fully reduplicated forms existed independently for any root with more than two consonants.

Finally, reduplication modifies the roots but is not inherently part of them. It is therefore important to keep this level distinct, including from a terminological perspective. To achieve this, I introduce the term BASE to refer to the reduplicated or non-reduplicated manifestation of a root. BASES are denoted by the symbol ⇒. BASES can be derived from synchronic roots, in which case they are synchronic bases, or from etymological roots, in which case they represent unattested proto-bases that can be assumed to predate the evidence from which they are inferred.

↳*wn*^{ME} : '(to) walk' = 2-SyR
⇒ $^{(-R)}$*wn* : '(to) walk'
⇒ $^{(+R)}$*wn* = *wnwn* : '(to) walk to and fro'

In this example, $^{(-R)}$*wn* is the non-reduplicated base, while $^{(+R)}$*wn* = *wnwn* is the reduplicated one.

5.3 Root Modifiers

5.3.1 Verbal Stems

In Egyptian/Coptic, the basic meaning of verbs can be modified through the systematic addition of prefixes to their consonantal skeleton. One prominent example is provided by the Early Egyptian prefix *s*- which turns the verb it combines with into its causative counterpart. Thus, *ḫpr* (inf.) : 'to happen', 'to become' serves as the starting point from which *sḫpr* (inf) : 'to cause to become' or 'to create' is derived. This type of derivation is very common and widespread in Semitic languages, where the resulting distinct configurations of the verb are usually conventionally called STEMS; I adopt the same term here. I refer to the unmarked form of a verb as the BASIC STEM, while all other forms are collectively termed DERIVED STEMS.

Stems are always derived from a BASE, which can be non-reduplicated or reduplicated, which in turn is derived from either an ETYMOLOGICAL or a SYNCHRONIC ROOT.

Occasionally, new stems can emerge in the course of the history of the language – for instance, Coptic causatives in *t*- can be synchronically analysed as a derived stem that emerged from the progressive lexicalization of the verb *(r)d* in pre-Coptic periphrastic causatives (see Kilani 2024). If the stem of a verbal form needs to be indicated explicitly, I suggest marking it by placing a superscript letter enclosed in square brackets before the form itself: $^{[B]}$ = basic stem, $^{[S]}$ = causative in *s*-, $^{[T]}$ = causative in *t*-; e.g. $^{[S](-R)}$*sw3d* = causative s-stem from the non-reduplicated base of the etymological root √*w3d*.

It is important to mention here that stems play a crucial role in understanding the diachronic development of the Egyptian verbal system, as they serve as the primary source from which new synchronic roots emerge. This process can be described as the progressive lexicalization of earlier stems of single verbs into new independent verbs over time. For example, in Old Egyptian, the non-reduplicated base of the synchronic root ↳*ḥ*ʿ^{OE} : '(to) stand' could take the form of both a basic stem $^{[B](-R)}$*ḥ*ʿ : '(to) stand' and a causative stem $^{[S](-R)}$*sḥ*ʿ : '(to) cause to stand'. However, by the time of Late Egyptian, the *s*- prefix was no

longer productive, and therefore these earlier stems had evolved into, and are thus better understood as, distinct synchronic roots.

These diachronic relations can be explicitly indicated with the notation introduced here, with the addition of the sign ➢ to indicate a stem derived from a base:

√**wrq*$^{\text{pr-prtEgy}}$ = 3-EtR
>√*wꜣḏ*^{prtEgy} = 3-EtR

↳*wꜣḏ*^{OE} : ‘(to) be(come) fresh/green’ = 3-SyR
⇒ $^{(\text{-R})}$*wꜣḏ*^{OE} : ‘(to) be(come) fresh/green’
➢ $^{[\text{B}](\text{-R})}$*wꜣḏ*^{OE} : ‘(to) be(come) fresh/green’
➢ $^{[\text{S}](\text{-R})}$*swꜣḏ*^{OE} : ‘(to) cause to be(come) fresh/green’

↳*sw{ꜣ}ḏ/t*^{LE} : ‘(to) refresh’ = 3/4-SyR[8]
⇒ $^{(\text{-R})}$*sw{ꜣ}ḏ/t*^{LE} : ‘(to) refresh’
➢ $^{[\text{B}](\text{-R})}$*sw{ꜣ}ḏ/t*^{LE} : ‘(to) refresh’

>↳*swt*^{DEM} : ‘(to) refresh’ = 3-SyR
⇒ $^{(\text{-R})}$*swt*^{DEM} : ‘(to) refresh’
➢ $^{[\text{B}](\text{-R})}$*swt*^{DEM} : ‘(to) refresh’

>↳*w{ꜣ}ḏ/t*^{LE} : ‘(to) be(come) fresh/green’ = 2/3-SyR
⇒ $^{(\text{-R})}$*w{ꜣ}ḏ/t*^{LE} : ‘(to) be(come) fresh/green’
➢ $^{[\text{B}](\text{-R})}$*w{ꜣ}ḏ/t*^{LE} : ‘(to) be(come) fresh/green’

⇒ $^{(\text{+R})}$*w{ꜣ}ḏ/t*^{LE} = *w{ꜣ}ḏ/tw{ꜣ}ḏ/t*^{LE}: ‘(to) be(come) (very?) green’
➢ $^{[\text{B}](\text{+R})}$*w{ꜣ}ḏ/tw{ꜣ}ḏ/t*^{LE} : ‘(to) be(come) (very?) green’

↳*wtwt*^{DEM} : ‘(to) be(come) (very?) green’ = 4-SyR
⇒ $^{(\text{-R})}$*wtwt*^{DEM} : ‘(to) be(come) (very?) green’
➢ $^{[\text{B}](\text{-R})}$*wtwt*^{DEM} : ‘(to) be(come) (very?) green’

>↳*wt*^{DEM} : ‘(to) be(come) fresh/green’ = 2-SyR
⇒ $^{(\text{-R})}$*wt*^{DEM} : ‘(to) be(come) fresh/green’
➢ $^{[\text{B}](\text{-R})}$*wt*^{DEM} : ‘(to) be(come) fresh/green’

5.3.1.1 Stem Morphemes – An Overview

Three main stem morphemes[9] can be identified throughout the history of the language. While the first two have good Afro-Asiatic parallels and therefore are likely part of a same single original derivational system, the last one is a later innovation that is fully manifest only in Coptic.

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8 The notation 3/4-SyR reflect the uncertainty about the status of *ꜣ* in this phase of the language.

9 Naturally, these are not the only prefixes or affixed morphemes attested in Egyptian (see, e.g., the instrumental/locative prefix m-). They are, however, the main ones that interact with verbal roots to form new verbal stems.

s- prefix

The first stem morpheme is the prefix *s*-, which was used to form the causative counterpart of a verb (see Schenkel 1999 for an overview; Štubňová 2019). This morpheme is uncontroversial and widely recognized and included in all major Egyptological grammars, although causative verbs are often categorized as "verbal classes" rather than as "stems". The Egyptian causative stem prefix also finds strong parallels in other Afroasiatic language families.

n- prefix

Several scholars have acknowledged the existence of an *n*- stem prefix in Egyptian (see Štubňová Nigrelli 2022 for a recent overview and discussion, with refs). However, this stem morpheme was already quite rare in Old Egyptian and is generally believed to have become non-productive by that time. Consequently, it is seldom discussed in Egyptological grammars. More often, this morpheme is simply treated as a root consonant, and verbs displaying it are frequently regarded as deriving from distinct roots that begin with *n*. The function of this prefix has been the object of some debate, but scholars seem to agree it may have originally somehow worked as a valency-decreasing prefix, either as a passive voice or anticausative. The prefix is likely connected to similar verbal morphemes found in other Afro-Asiatic language families.

t- prefix

Coptic attests the existence of a pre-Coptic stem morpheme *t*- (ⲧ- |t-| in all dialects) originally used to form causative verbs. This morpheme does not appear to be productive anymore in Coptic and only appears as a lexicalized element in specific verbs. The etymology of this prefix is clear: it is the result of the lexicalization of the light causative verb *(r)d* : '(to) give', '(to) place'. However, in spite of this verbal origin, it is important to point out here that by the time of Coptic this morpheme had become a real stem morpheme, as verbs displaying it behave as fully autonomous verbs that can be regularly conjugated (e.g. in the stative), something that would not be possible if this were still a [auxiliary_verb + lexical_verb] construction (see Kilani 2024 for a recent discussion).

5.4 Morphological Modifiers

5.4.1 Vocalization: Proto-Coptic **a* vs **E*

Pre-Coptic forms of Egyptian were written using only consonants. As a result, the vocalization of Egyptian verbs, along with any morphological patterns or oppositions that may have characterized them, can only be investigated through its later stage, Coptic, where these features were at least partially preserved. Therefore, from a conceptual perspective, the opposition discussed here, like most of those examined in the following sections, is more accurately described as generically proto-Coptic, rather than as specifically pertaining to any attested form of pre-Coptic Egyptian. This is because these features can only be studied through reconstruction from Coptic and are not directly observable in any of the

attested forms of earlier stages of the language. Nonetheless, it is very likely that they already existed and influenced verbal morphology during those earlier phases. However, correlating these reconstructed proto-Coptic patterns with the attested pre-Coptic phases of the language lies beyond the scope of this paper and must be left to future research.

Unfortunately, the Coptic evidence also presents several major limitations. First, by the time of Coptic, unstressed vowels had been reduced to an indistinct vowel (possibly [ə] or the like), making them largely impossible to reconstruct. Occasionally, though, these vowels can be recovered from secondary verbal forms where the stress shifted to a syllable that is typically unstressed (e.g., prepronominal forms in opposition to absolute forms). Second, it is evident that several verbal forms that existed in Egyptian are either poorly preserved in Coptic (e.g., attested only for certain types of verbs) or have not survived at all. This means that any analysis of Egyptian verbal vocalization is necessarily limited to the stressed vowels of those verbal forms that are attested in Coptic; anything else is beyond our reach.

Finally, the Egyptian vocalic system is typically reconstructed as having an opposition between three vowels: **a*, **i*, and **u*. This reconstruction is primarily modelled on the tri-vowel system proposed for Proto-Semitic (in line with the common assumption that the Egyptian system must have been similar to that of Semitic) and on a limited number of cuneiform transcriptions, which however should be regarded as inherently ambiguous.[10]

I am sceptical about this three-vowel model, and it seems to me that the evidence supports with any certainty the reconstruction, for proto-Coptic, of only a two-vowel opposition[11] between a vowel **a* and a vowel *non-*a*, which I conventionally transcribe as **E*.[12] This is not new: even the scholars who support an original tri-vowel system recognize that the supposed **i* and **u* vowels had merged in pre-Coptic time and are indistinguishable

10 Cuneiform was not designed to transcribe Egyptian vowels, and therefore these transcriptions should be understood as attempts to transcribe in cuneiform a foreign vocalic system (the Egyptian one) that might have been quite different from that of Akkadian. In this respect, it is worth noting that cuneiform often used signs with all three vowels <a>, <i>, <u> to transcribe foreign /ə/, as it lacked dedicated signs for that vowel. A good example of this phenomenon is provided by the variants *niq2-**me**-pa-a* / *niq2-**mu**-pa* / *niq2-**ma**-pa-a*, all attested in 13th-century Ugarit and understood as attempts to render an Amorite name that was likely pronounced *niq**mə**pa* at the time (Howard 2023: 50). In general, cuneiform is notoriously idiosyncratic in its transcription of foreign vowels, as illustrated by the remarkable variability in the rendering of foreign personal names (see Waerzeggers and Gross 2024). As far as we know, the same may well have been true for cuneiform transcriptions of Egyptian words, and the use of cuneiform <i> and <u> might reflect (allophonic?) approximations of a single vowel missing in the phonological inventory of cuneiform Akkadian, rather than a reliable indication of the existence of a phonemic /i/ and a phonemic /u/ in Egyptian.

11 It is interesting to mention here that a two-vowel system with an opposition **a* vs **ə* has been suggested for Proto-Afro-Asiatic at least once in the past (Diakonoff et al. 1993).

12 It is true that this **E* vowel appears in different forms in Coptic, some of which can clearly be explained as positional or contextual allophones, while a few remain unexplained. Supporters of the three-vowel model have attempted to derive these unexplained variants of **E* from an earlier **i* vs. **u* opposition, but in my opinion, none of the proposed models is truly convincing, and it seems just as likely that these variants reflect additional allophonic variation caused by factors that are not

on the basis of Coptic alone.[13] A vocalic system based on a binary opposition becomes particularly appealing when we consider that the proto-Coptic verbal system as it can be reconstructed from Coptic consistently exhibits a broad contrast between two vowels (**a* and **E*), with no solid evidence for a morphological three-way opposition requiring a third distinct vowel (see below). Moreover, in my view, this strongly suggests that this two-vowel proto-Coptic system might actually be quite old, possibly going back as early as the proto-Egyptian stage, when the verbal system attested in the historical phases of the language was formed. However, whether this is ultimately true or not is not that relevant for this paper, as the focus here is on discussing the patterns and oppositions that are attested.

This vocalic opposition between **a* and **E* appears to have been central to the pre-/proto-Coptic verbal system, and thus deserve to be discussed in detail here.

We can start by saying that while for several verbs and verbal forms only a vocalization in **a* or one in **E* is attested, a few verbs attest both vocalizations in the same verbal forms. In turn, the fact that a same verbal form could display both vocalizations suggests that this opposition was not a mere lexical phenomenon, but encoded some semantic or morphological property. Moreover, the opposition was clearly not limited to a single verbal form, as it can be observed across different ones. For instance, the verb [S] ϣτοϩ |šᵊtɔ́m|[14] / [S] ϣταϩ |šᵊtám| : 'to shut', 'to be shut' (intransitive) display two infinitives that are identical except for the stressed vowel: in the first, [S] ϣτοϩ |šᵊtɔ́m| < **ḫVtám*, the stressed vowel derives from a stressed **a*, while in the second, [S] ϣταϩ |šᵊtám| < **ḫVtÉm*, it derives from a stressed **E*. Similar alternations are attested also in the pseudoparticiple/stative; see e.g. ϫροειτ |čᵊrɔ́jt| < **ḏVráỷ-tV* vs ϫραειτ |čᵊrájt| < **ḏVrÉỷ-tV* : 'being strong/victorious' (see note 18 for examples). One can also mention here a few verbs that present two pseudoparticiples/statives, one characterized by a stressed [S] ο |ɔ| < **a* in closed syllable and the other by a stressed ⲏ |e| < **E* in open syllable, although here there is also an apparent difference in syllabic structure;[15] see e.g. [S] ποσε |pɔ́sə| vs [S] ⲡⲏⲥ |pés| : 'being cooked'.

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yet understood. All of this, however, requires further investigation that falls beyond the scope of this paper.

13 This is clear in closed syllables (see e.g. Osing 1976: 11 and passim). In open syllable some scholars have tried to suggest rules to distinguish them (see e.g. Osing 1976: 19 and passim), but on the one hand these rules are far from being systematic and on the other they are full of exceptions in which the supposed **u* appears to behave like the supposed **i*, and the supposed **i* appears to behave like the supposed **u*, to the point that it is essentially impossible to predict the supposed original vowel on the basis of Egyptian evidence alone (see e.g. Osing 1976: 19 and passim; or Peust 1992 for a good example of how the difference in vocalization displayed by Coptic might actually be explained as allophonic variants of one single protovowel).

14 I will refrain from discussing here verbs containing the consonants *ʾ* and *ʿ*, unless they provide the only or clearest example of a specific phenomenon. These consonants undergo distinctive developments from Egyptian to Coptic and interact in complex ways with neighboring consonants, requiring detailed explanations that are not directly relevant to the main arguments of this paper.

15 These are primarily (although not exclusively) verbs classified as 2-lit in the traditional system. It is intriguing to consider whether the form in ο |ɔ| might derive from an original structure **CaXVCV*

The grammatical or semantic difference that must have been encoded by this vocalic opposition has received very little attention from scholars. Vycichl associated **E* with a passive meaning, particularly in relation to participles (see e.g. Vycichl 1959). However, while a plain active vs. passive distinction might explain some of the attestations, it cannot account for all instances of this opposition. For example, the infinitive form of the verbs [S] ⲥⲛⲁⲧ |sˁnát| < **sVnÉḏ* : 'to fear', 'to be afraid' and [S] ϭⲛⲁⲧ |kʲˁnát| < **qVnÉd* 'to be angry' cannot be easily interpreted as being passive.

I propose an alternative and more nuanced interpretation based on another set of verbal forms that consistently exhibit both vocalizations: the infinitives of so-called reduplicated verbs. Coptic features a substantial number of reduplicated verbs derived from both 2-SyR^{prtCpt} and 3-SyR^{prtCpt} roots. While the majority of these verbs display a vocalization in **a*, a significant subset showcases a vocalization in **E*. Table 1 illustrates examples of both types.

Table 1 | Examples of 2-SyR^{prtCpt} and 3-SyR^{prtCpt} with vocalization in **a* and **E*.

2-SyR^{prtCpt} – *C^{1}áC^{2}C^{1}VC^2	2-SyR^{prtCpt} – *C^{1}ÉC^{2}C^{1}VC^2
[S] ϣⲟⲧϣⲧ |šótšˁt| 'to cut, carve, hollow'	[S] ⲃⲉⲉⲃⲉ |bɛ́ːbə| 'to bubble up', 'to swell', 'to leak'
[S] ⲧⲟϭⲧϭ |tɔ́kʲtˁkʲ| 'to press firmly'	[S] ⲧⲏⲧⲙ |tń̥tm̥| 'to be oppressed', 'to be heavy'
[B] ϑⲟⲩⲧⲉⲫ |tʰɔ́ftˁf| 'to let fall drop by drop (trans.)'	[S] ⲧⲗⲧⲗ |tĺ̥tl̥| 'to drip'

3-SyR^{prtCpt} – *C^{1}VC^{2}áC^{3}C^{2}VC^3	3-SyR^{prtCpt} – *C^{1}VC^{2}ÉC^{3}C^{2}VC^3
[S] ⲕⲗⲟⲙⲗⲙ |klɔ́ml̥m| 'to embrace', 'to be twisted'	[S] ⲕⲣⲙⲣⲙ |kŕ̥mr̥m| 'to be vexed', 'to murmur'
[A2] ϫⲣⲁⲣⲉ |črárə| 'to scatter, disperse (trans)'	[F] ⲥⲗⲉⲍⲗⲉϭ |slékʲləkʲ| 'to be rolled', 'to roll'
[S] ϩⲣⲟϫⲣⲉϫ |hrɔ́črəč| 'to grind, rub', 'to be set on edge'	[S] ⲥⲛⲁⲉⲓⲛ |snájn(ˁ)| 'to stroll, wander around'

or **CaCVXV*, analogous to the **CaCVCV* structure of *3-SyR* verbs, where *X* was a consonant that did not appear in writing (possibly because somehow akin to a glide that might have been somehow assimilated to the nearby vowels?). This could have undergone a development like **CaXVCV* > **CaXCV* or **CaCVXV* > **CaCXV*. Meanwhile, the form with ⲏ |e| might stem from a structure **CEXVCV* > **CEXCV* or **CECVXV* > **CECXV*, paralleling the **CECVCV* structure rarely displayed by the stative of some 3-SyR verbs (cf. [B] ϧⲉⲛⲧ | hɛ́nt| < **ḫÉndV* : 'approached'). In this case, the sequence **-EXC-* or **-ECX-* could have been assimilated into a single segment **-E-*. If this is the case, then it might suggest the hypothetical consonant *X* was phonologically close enough to **E* to be assimilated by it, while distinct enough from **a* to maintain its function as consonantal coda of the syllable. This hypothesis is, of course, highly speculative.

What is evident from this small selection is that the opposition between **a* and **E* is not strictly one of active versus passive. Yet, agentivity does appear to play a role in the distribution of these vowels. In particular, verbs with **a* clearly convey active meanings, describing actions where the (implicit) main argument[16] is the main agent or performer. In contrast, verbs with **E* seem to describe actions where the (implicit) main argument functions more as a non-actant experiencer or lacks full or direct control over the action.[17] This is particularly clear in the pair [B] ϭⲟⲩⲧⲉⲩ |tʰɔ́fᵊf| : 'to let fall drop by drop (trans.)' vs [S] ⲧⲁⲧⲗ |tḭ́tḭ| : 'to drip'. The first verb, with **a*, focuses on the action actively performed by someone to cause a liquid to fall drop by drop, with the agent being the main participant of the action. In contrast, the latter verb, with **E*, describes the dripping of liquid, with the liquid itself as the main, passive participant of the action.

The vowel **E* would thus correspond to marked situations where the verb describes an action or state over which its main participant has no direct control, i.e. action with low internal agentivity. These forms typically reflect scenarios in which the main participant lacks full autonomy or functions as a passive experiencer of the process, which, in turn, can be assumed either to take place on its own, without external triggers, or to be to result of the actions of an implicit or explicit external agent, which however is not the main participant of the action. Forms with **E* could then be seen as passive, resultative, or expressing spontaneous/self-contained processes.

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16 These forms are infinitives, so technically they can exist without arguments. Yet from a semantic point of view, the action described by the infinitive can also have an implicit argument: 'to go' implies that something/someone can perform the action of going, 'to drop' implies that something/someone can drop something, etc.

17 Interestingly, there is a consistent number of reduplicated verbs in **E* that belong to the very specific semantic field of verbs describing the production of sound; e.g. [S] ⲥⲛⲥⲛ |sṅ̓sn̓| = 'to resound', [B] ⲃⲉϧⲃⲉϧ |bɛ́hbəh| = 'to howl', 'to bark'; [S] ϫⲓⲛϫⲓⲛ |čínčin| = 'to make music (with instrument or voice)'. I see two possible, although very tentative explanations of this phenomenon. First, it is possible that Egyptians conceptualized the production of sound as something that is experienced by the argument, rather than actively performed by it. This point of view is certainly easy to understand in the case of musical instruments, as they produce sounds not autonomously, but because played by a musician = an external agent. However, it is admittedly harder to see how this could apply to the production of sound by living entities, like in the case of [B] ⲃⲉϧⲃⲉϧ |bɛ́hbəh| = 'to howl', 'to bark'; perhaps there as well the idea could be that the sound-producing organs are not active agents, but rather experiencers of the will of the entity (human or animal) using them? The other possible explanation stems from the observation that Berber has a whole series of expressive/idiophonic/onomatopoeic verbs characterized by partial or full reduplication of the root and by a vocalization comparable to Egyptian **E* (Dingemanse 2008; see also van Putten 2024: passim). Perhaps this similarity is not accidental: perhaps these Egyptian-Coptic sound-related verbs should also be understood as inherently expressive, idiophonic, or onomatopoeic, and this peculiar nature might be one more semantic nuance associated with the non-actant vowel **E*. Alternatively, we might even consider a more direct connection: most if not all of the Egyptian reduplicated verbs associated with sound are not attested before Demotic or at most Late Egyptian. This could suggest that the association of these idiophonic/sound-related verbs with reduplicated forms vocalized in **E* might actually be a secondary, and rather late development, and one may wonder whether the Libyan presence in the Late Period might have played a role in their emergence.

This contrasts with verbal forms in **a*, which appears to correspond to situations implying high internal agentivity, that is, where the action is actively performed by the main participant (as with dynamic transitive and intransitive verbs), or where no process/no change of state is involved at all (as with non-dynamic quality verbs).

For instance, the pseudoparticiple in **a* ϫⲣⲟⲉⲓⲧ |čˢrə́jt| < **ḏVráẏ-tV* might represent an agentless non-dynamic stative 'being (inherently) strong/victorious', while that in **E* ϫⲣⲁⲉⲓⲧ |čˢrájt| < **ḏVrÉẏ-tV* could indicate a dynamic resultative 'having become strong/victorious (possibly due to an external agent/instrument)'.[18]

Similarly, the opposition between the infinitives [S] ϣⲧⲟⲙ |šˢtə́m| < **ḫVtám* and [S] ϣⲧⲁⲙ |šˢtám| < **ḫVtÉm*, both seemingly meaning 'to shut', 'to be shut' (intransitive), could reflect different levels of agency: [S] ϣⲧⲟⲙ |šˢtə́m| < **a* might mean 'to be (actively or statively) a closure', while [S] ϣⲧⲁⲙ |šˢtám| < **E* could have originally meant 'to be a closure (because of someone's actions)'; see below for further discussion and examples (§5.4.2.1).

The vowel **E* in [S] ⲥⲛⲁⲧ |sˢnát| < **sVnÉḏ* : 'to fear', 'to be afraid' and [S] ⲕⲛⲁⲧ |kʲˢnát| < **qVnÉd* 'to be angry' may also align with this interpretation, reflecting a condition of being afraid/angry (i.e. passive experiencer) caused by an external agent triggering the fear/anger.

5.4.2 Vocalization: Stress Patterns in Infinitives/Action Nouns

In addition to the opposition between the two pre-Coptic vowels **a* and **E*, the Coptic verbal system also exhibits a contrast in the position of the stressed vowel, either toward the beginning or the end of the word. This positional opposition is observed in both infinitives/verbal nouns and participles/actant nouns.

The fact that forms with both prosodic patterns are attested for the same verbs suggests that the opposition is morphologically significant. However, it remains unclear whether the morphological distinction it encodes in infinitives/verbal nouns is the same as that found in participles/actant nouns.

Accordingly, the following paragraphs will assume the two phenomena as distinct and independent, and will focus on the infinitives/verbal nouns. The study of participles/actant nouns will not be attempted here, as it would require a discussion that is beyond the possibilities of this article.[19] Consequently, also the possibility of a deeper connection

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18 A quick look at the Coptic attestations might support this suggestion. See for instance (examples from the Annis database https://corpus-tools.org/annis/):
1. [S] ⲛⲓⲙ ⲡⲉⲧϫⲣⲟⲉⲓⲧ ⲉⲡⲕⲟⲥⲙⲟⲥ : "Who is he who is (inherently) victorious (non-dynamic stative) over the world" (sahidica.nt > 62_1_John_05 : norm_group 52 - 62)
versus:
2. [S] ⲁⲡⲁ ⲃⲓⲕⲧⲱⲣ ⲛⲉⲧϫⲣⲁⲉⲓⲧ ⲡⲉ ⲉⲙⲡⲉϣⲗⲏⲗ : "Apa Victor had become victorious (dynamic resultative) over him by means of prayers" (martyrdom.victor > martyrdom.victor.02 : norm_group 17 - 27).

19 The situation with actant nouns is generally hard to evaluate. While many attested forms exist in Coptic, most are lexicalized and often morphogrammatically opaque. Additionally, actant nouns often have an additional suffix *-ẏ*, whose function is unclear, which may or may not be related with

between this prosodic opposition in infinitives/verbal nouns vs in participles/actant nouns will not be discussed here.

5.4.2.1 Infinitives / Verbal Nouns

This opposition can be described as follows: in the infinitives of 3-EtR^{prtCpt} verbs, the stressed vowel can occupy one of two positions, namely either between the first and second consonants or between the second and third consonants; for example, [S] ϣⲱⲧⲙ |šótᵊm| : 'to shut (something)' vs. [S] ϣⲧⲟⲙ |šᵊtóm| : 'to shut', 'to be shut' vs [S] ϣⲧⲁⲙ |šᵊtám| : 'to shut', 'to be shut'. Similar oppositions can be found in other types of infinitives (e.g. [A] ⲥⲟⲟⲩⲉ |sóːhə| vs [A] ⲥⲉⲩⲟ |səhó| : 'to set up', 'to cause to stand').

The underlying phonological mechanics of this phenomenon have received little attention. To my knowledge, only Vycichl (see e.g. 1984, X) provides an explanation. In particular, building on Fecht's syllabic theory, he proposes that by default, Coptic stress falls onto the second-to-last syllable. This syllable is typically open in words with three consonants or two consonants followed by a final glide (e.g., *CV́.CVC|ỷ*), while it is closed in words with more than three consonants or more than two consonants plus a final glide (e.g., *CV́C.CVC|ỷ*, *CV.CV́C.CVC|ỷ*, etc.). The latter may further result from an earlier syncope of a medial vowel in earlier **CV́.CV.CVC|ỷ* et sim. patterns (i.e., **CV́.CV.CVC|ỷ* > *CV́C.CVC|ỷ*). Words deviating from these patterns fall into two main groups. On the one hand, there are forms with more than three consonants or more than two consonants and a final glide, in which the stressed syllable is open. Vycichl explains these cases as either resulting from the simplification of a consonant cluster or a geminate consonant following the stressed vowel (i.e. *CV.CV́.CVC|ỷ* < **CV.CV́C.CVC|ỷ* / **CV.CV́C.ːVC|ỷ*), or from the presence of an etymologically long proto vowel in the stressed syllable, which inhibited the syncope of the penultimate vowel (i.e. *CV.CV́.CVC|ỷ* < **CV.CV̋.CVC|ỷ*). On the other hand, there are words where the stress falls on the final syllable. Vycichl attributes this pattern to the presence of an etymologically long proto vowel in the final syllable, which prevented the stress from shifting further back (i.e. *CV.CV́C|ỷ* < **CV.CV̋C|ỷ*).

Within this framework, Coptic infinitives that conform to the default syllabic rule can be assumed to have had no etymological long vowel (e.g. S ϣⲱⲧⲙ |šótᵊm| < **ḫát(V)m*), while those that do not conform to it can be assumed to have had an etymological long vowel in their final syllables (e.g. ϣⲧⲟⲙ |šᵊtɔ́m| < **ḫVtā́m*).

This interpretation not only provides an elegant explanation for the stress placement in these Coptic forms but also highlights compelling parallels with Semitic. Specifically, as Vycichl has already observed, the main infinitive patterns attested in Coptic can be reconstructed as [S] CѡC(ϵ)C |CóC(ə)C| < **CáC(V)C*, [S] CıCCϵ |CiCC-ə| possibly < **CÉC(V)C-Vt* (if the analysis presented in §5.3.3.2 is correct), [S] C(ϵ)CoC |C(ə)CɔC| <

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the *-ỷ* of infinitives (see below), and which may have altered the prosodic structure of the words, making it difficult to determine what the original vocalization actually was. Things are even more complex with roots that are not 3-EtR^{prtCpt}, as even more interpretations become possible. All these issues warrant dedicated studies, which would go beyond the scope and reasonable length of a paper like the present one.

**CVCáC*, [S] C(ɛ)CꜣC |C(ə)CáC| < **CVCÉ́C*, which are all attested in Semitic as well-documented patterns for infinitives or action nouns. I find this analysis reasonable (and I am not aware of more convincing explanations); I suggest to represent this opposition as -V̄ vs +V̄, i.e. 'without long vowel' vs 'with long vowel'.

The fact that certain verbs have infinitives displaying both types of patterns (e.g. as mentioned above -V̄ [S] ⱷⱳⱅⲙ |šótˢm| vs +V̄ [S] ⱷⱅⲟⲙ |šˢtóm|) clearly indicates that these forms are not mere lexical variants but must instead encode some form of morphosemantic difference.

Defining the precise nature of this distinction remains challenging, not least because these forms have received scant scholarly attention. A reassessment of the evidence is therefore in order, beginning with a clearer delineation of the forms in question.

When 3-EtR^{prtCpt} verbs are considered, four groups of verbs appear to deviate from the default vocalization patterns expected on the basis of the standard syllabic theory and are therefore likely to reflect a +V̄ structure:

1. The first group includes verbs that, in Sahidic, display a vowel such as ⲟ |ɔ| < **a* or ⲉ/ⲁ |ɛ|/|a| < **E* in their final syllable. These vowels typically occur in closed syllables (CVC), suggesting an underlying original pattern of the type **CVCV́C*. E.g. [S] ⱷⱅⲟⲙ |šˢtóm| : 'to shut', 'to be shut'; [S] ⲥⲛⲁⲧ |sˢnát| : 'to fear', 'to be afraid'.
2. The second group consists of verbs in which the stressed vowel in Sahidic is a ⲱ |o| < **a* or ⲏ |e| < **E*, both vowels typically found in open syllables (CV). These forms point to an original structure **CVCV́CVẏ*.
   E.g. [S] ⲟⲣⲱϩ |kʲˢróh(ˀ)| = 'to be in want', 'to be needy', 'to be diminished'; [S] ⲗϩⲏⲙ |lˢhém(ˀ)| : 'to roar'.
3. A very small number of causative verbs also show stress on the final syllable, with some presenting a vowel characteristic of a closed syllable (especially ⲟ |ɔ|, no form with ⲉ/ⲁ |ɛ|/|a| is known to me), implying an original pattern such as **sVCVCV́C*.
   E.g. [A] ⲥⲉϩⲟ |səhɔ́| : 'to set up', 'to cause to stand'.
4. Finally, a couple of causative verbs display a stressed vowel typical of an open syllable (especially ⲏ |e|, no case with ⲱ |o| is known to me), suggesting an original pattern **sVCVCV́CVẏ*.
   E.g. [A] ⲥⲟⲣⲏϩ |sˢkʲréh(ˀ)|: 'to rest', 'to pause', 'to be quiet', 'to rest oneself'.

Verbs in group 1 can be interpreted as base stem forms and will serve as the foundation for the following discussion. By contrast, those in groups 2, 3, and 4 likely belong to derived stems (see above §5.3.1.1) or stem extended with a suffix (see below §5.4.3.1) and will be set aside for now.

As for other verbal classes, the situation is more complex. First of all, it is important to notice that the most common pattern for infinitives of so-called 2-lit verbs is [S] CⱱC |CóC|, and it is reasonable to assume that this form is the equivalent of the [S] CⱱC(ɛ)C |CóC(ə)C| < **CáC(V)C* of so-called 3-lit verbs. Yet, the form [S] CⱱC |CɔC| is somehow

problematic, as if the original pattern was just **CáC*, we would expect [S] ⲟ |ɔ|, not [S] ⲱ |ó|. Some scholars (see e.g. Fecht 1960: 3, Osing 1976: 36, Vycichl 1984: 324) have argued that this is an irregular innovation modelled on the infinitives of 3-lit verbs, but I find this explanation rather *ad hoc* and ultimately unlikely, as there is no evidence this form is in anyway 'newer' than other forms. Rather, a phonological explanation seems more likely to me. One option is to assume that these verbs originally had a third undetermined but likely weak consonant (I indicate it with *Χ*), either in their middle or at their end, that was generally not spelled out in hieroglyphs but which did affect their syllabic structure and overall vocalization. In this case, we could assume that [S] CⲱC |CóC| < **CáΧ(V)C / *CáC(V)Χ*; this would have a good parallel in the case of verbs with a middle consonant *ꜣ*, which display exactly the same development, e.g. [S] ϩⲱⲃ |hób| < **háꜣ(V)b* : 'to send'. A particularly intriguing piece of evidence supporting this hypothesis comes from the verb 'to die': [S] ⲙⲟⲩ |mú|, which is consistently written as *mt* (with only two consonants) in pre-Coptic Egyptian, and whose Coptic infinitive follows the regular CⲱC pattern.[20] Yet the verb is generally traced back to a proto-Afroasiatic root reconstructed as √*mwt*, with **-w-* (or the like) as the second consonant (cf. **mawVt-* in Orel and Stolbova 1998). It is tempting to see a connection between this Afroasiatic **-w-* and the hypothetical consonant *-Χ-* proposed here. Additional elements in support of this interpretation will emerge in the following sections of this article.[21] Nevertheless, it is important to stress that the presence and possible location in the root of this third consonant *-Χ-* remain hypothetical, as only the two strong consonants are usually attested in writing. For this reason, it is better to refer to these roots as 2⁺-EtR^{prtCpt}, i.e. as roots with *at least* two consonants.

Besides the 2⁺-EtR^{prtCpt} just discussed, the pattern [S] CⲱC |CóC| is also sporadically attested for infinitives of verbs traditionally classified as II-gem and III-weak; these instances are further discussed below.

As for potential +V̄ infinitives of non-3-EtR^{prtCpt} verbs, one can first observe that there are no clear examples of such patterns among verbs traditionally classified as 2-lit or III-weak. Osing (1976, 44–45, 60–62) proposes that a very small number of forms, which he derives from patterns he reconstructs as **CaC* and **CVCEt*, might fall into this category. However, these cases are all problematic and ultimately, in my opinion, unconvincing—either because the roots include consonants not preserved in Coptic, because the verbs may in fact be 3-EtR^{prtCpt}, or because the roots are unattested in earlier stages of Egyptian. By contrast, there are quite a few examples of verbs traditionally classified as II-geminate that display infinitives clearly reflecting +V̄ patterns, but in the form **C^{1}VC^{2}V́C^{2}* , i.e. through a pattern characterized by the reduplication of the second consonant (e.g. [S] ⲕⲙⲟⲙ |kᵊmóm| : 'to be(come) black').

This distribution is undeniably striking and intriguing, and it brings us back to the core issue of verbal classification discussed in this paper. If the traditional verbal classes were valid, we would expect to find at least some reliable examples of verbs displaying +V̄ patterns in each of them. Instead, such forms are common in the so-called 3-lit and II-gem

20 With the regular ⲱ |o| > ⲟⲩ |u| after nasal and loss of final *-t#*.

21 See also footnote 15 above.

classes, but are notably absent from the so-called 2-lit and III-weak classes. However, this apparent anomaly disappears if we move away from the traditional Egyptological verbal classes, as this would open a new intriguing possibility: what if we have +V̄ patterns for so-called II-gem verbs, but not 2-lit and III-weak verbs, because in fact these are not distinct verbal classes, but simply distinct paradigmatic manifestations of one single 2⁺-EtR^{prtCpt} verbal class?

This is the primary hypothesis that will be explored in the remaining part of this article. Compelling support for this hypothesis is provided by Coptic. In particular, it is remarkable that some verbs traditionally classified as II-gem and reflecting a *$C^1VC^2\acute{V}C^2$ infinitive also show an alternative infinitive with the [S] CѡC pattern, namely the pattern that is characteristic of so-called 2-lit verbs;[22] for example, [S] ϩѡⲛ |hón| < *$\underline{h}$*án(V)* vs [Sa] ϩⲛⲁⲛ |hˢnán| < *$\underline{h}$*Vnā́n* : ‘to approach’, ‘to be near’.

It is worth noticing here that most verbs reflecting a *$C^1VC^2\acute{V}C^2$ pattern are characterized by a vowel **a*. I am aware of only a few verbs that presents this pattern with a vowel **E*, such as [S] ⲧⲗⲗ |tˢlél| < *$\underline{t}$*VrÉ̄r* : ‘to rejoice’. The semantics of this verb align well with the meaning of the vowel **E* argued above, as ‘to rejoice’ can be understood as having low agentivity and being triggered by an external agent, namely: ‘to rejoice’ = ‘to experience happiness (because of some external event)’.

The basic paradigm of this new single 2⁺-EtR^{prtCpt} class would thus look as in table 2 – this is the model followed in this article.

Table 2 | **CáC(V)C*, **CVCáC*, and * *CVCÉ̄C* in 3-EtR^{prtCpt} and 2⁺-EtR^{prtCpt} verbs.

BASE:	**CáC(V)C*	**CVCáC*	**CVCÉ̄C*
3-EtR^{prtCpt}	**CáC(V)C*	**CVCáC*	**CVCÉ̄C*
$\underline{h}$tm	[S] ϣⲱⲧⲙ |šótˢm| < **$\underline{h}$*át(V)m*	[S] ϣⲧⲟⲙ |šˢtóm| < **$\underline{h}$*Vtā́m*	[S] ϣⲧⲁⲙ |šˢtám| < **$\underline{h}$*VtÉ̄m*

2⁺-EtR^{prtCpt}	*C^1*á(XV)C* or *C^1*áC(VX)²*	*C^1VC^2*áC²*	*C^1VC^2*É̄C²*
$\underline{h}$n	[S] ϩѡⲛ |hón| < **$\underline{h}$*á(XV)n* or **$\underline{h}$*án(VX)*	[Sa] ϩⲛⲁⲛ |hˢnán| < **$\underline{h}$*Vnā́n*	
$\underline{t}$r			[S] ⲧⲗⲗ |tˢlél| < **$\underline{t}$*VrÉ̄r*

While interpreting so-called II-geminate verbs as just a manifestation of a common 2⁺-EtR^{prtCpt} class significantly simplifies the overall picture, the form of their +V̄ infinitives raises an important question: what is the origin of the reduplication of the second consonant?

I see three possible explanations:

22 But which are also attested for certain III-weak verbs, as it will be discussed below.

1. The reduplicated consonant is part of the root. This would imply that:
   a. These forms are non-reduplicated B-stem, as are the corresponding 3-EtR^{prtCpt} forms.
   b. 2^{+}-EtR^{prtCpt} verbs ultimately derive from pre-proto-Coptic etymological 3-consonant roots with a reduplicated second consonant (i.e. 3-EtRpr-^{prtCpt} with II=III).
   c. These reduplicated consonants are lost or invisible in most forms due to the absence of a stressed vowel between them and the general simplification of geminates in Egyptian.

2. The reduplicated consonant is not part of the root, nor is it a morphological reduplication. Instead, it is an additional "structural" consonant introduced as needed just to turn the consonantal skeletons of these verbs into three-consonant ones to accommodate this peculiar vocalization. This would mean that:
   a. These verbs are ultimately derived from roots with only two consonants, i.e. they are truly just 2-EtR^{prtCpt} roots.
   b. These forms are therefore regular B-stem, as are the corresponding 3-EtR^{prtCpt} forms, and the reduplication of the consonant has no meaning per se.

3. The reduplicated consonant represents a context-specific realization of the hypothetical consonant -$\mathcal{X}$- proposed for 2^{+}-EtR^{prtCpt} verbs. This would imply that:
   a. These verbs are ultimately derived from roots with three consonants, two strong ones and a third weak one that takes the form of the last strong one in +$\bar{V}$ forms, namely: *$C^1VX\acute{V}C^2$ or perhaps *$C^1VC^2\acute{V}\mathcal{X}$ > *$C^1VC^2\acute{V}C^2$.
   b. These forms are therefore regular B-stem, as are the corresponding 3-EtR^{prtCpt} forms, and the reduplication of the consonant has no meaning per se.

While at the moment no definitive evidence supports any of the three hypotheses, I find the third option more compelling: as discussed above, and as will be argued further below, positing the presence of a third consonant -$\mathcal{X}$- provides an efficient working solution for explaining several oddities across the system. For this reason, the third hypothesis will be adopted here as a working assumption. It must be stressed, however, that this remains provisional and subject to revision pending further evidence.

The next obvious question concerns the nature of the opposition between -$\bar{V}$ and +$\bar{V}$ infinitives. It is commonly assumed that intransitivity plays a role, as +$\bar{V}$ infinitives are often clearly intransitive. This explanation, however, seems unconvincing to me, as numerous -$\bar{V}$ infinitives are also intransitive.

Instead, I tentatively propose interpreting the -$\bar{V}$ / +$\bar{V}$ opposition as one of actionality or dynamicity. In this view, -$\bar{V}$ infinitives would describe the dynamic occurrence of a specific action (they would thus be real ACTION NOUNS), while +$\bar{V}$ infinitives would reflect ongoing but non-dynamic states (thus being more generic VERBAL NOUNS). This analysis is supported by the semantic peculiarities of the verbs displaying these infinitives. First,

most of the Coptic +V̄ B-stem infinitives derive from quality verbs,[23] which are inherently non-dynamic and can be described as STATE VERBAL NOUNS, in contrast to dynamic ACTION NOUNS; e.g. [S] ⲟⲩⲙⲟⲧ |wˢmɔ́t| : 'to (be)come thick', [S] ⲁⲣⲟϣ |arɔ́š| : 'to (be)come cold', [S] ⲕⲙⲟⲙ |kˢmɔ́m| : 'to be(come) black', [B] ϧⲃⲟⲃ |kˢbɔ́b| : 'to (be)come fresh', etc.

It is also important to emphasize that these qualities can be understood as ongoing or enduring (i.e. 'to *currently* be X'), rather than tied to a specific action as with -V̄ infinitives. This suggests that +V̄ forms may also carry a [+imperfective] aspect.

Furthermore, non-dynamic interpretations can also be applied to most, if not all, of the non-quality verbs that display +V̄ infinitives. For example, the -V̄ infinitive [S] ϣⲱⲧⲙ |šótˢm| could signify a dynamic 'to close' = 'the act(ion) of closing' = 'the act of turning something from open to closed'. By contrast, the +V̄ infinitive [S] ϣⲧⲟⲙ |šˢtɔ́m| < **ẖVtám* would correspond to a non-dynamic, ongoing 'to close' = 'to be closing' = 'the state of being closed/being something that closes'. This form would also imply an [actant]/[no external agent] context (expressed by the vowel **a*) and would primarily be intransitive. Lastly, the +V̄ infinitive ϣⲧⲁⲙ |šˢtám| < **ẖVtÉm* might represent a non-dynamic, resultative, ongoing 'to be closed' = 'to be closing due to someone's intervention' = 'the state of being closed/being something that closes due to someone's intervention'. This form would indicate a [non-actant]/[with external agent] context (vowel **E*) and would be primarily intransitive. All this can be summarized as in table 3.

Table 3 | Infinitives of the verb [S] ϣⲱⲧⲙ |šótˢm| analysed according to the -V̄/+V̄ and **a*/**E* oppositions.

CV́CC	ϣⲱⲧⲙ |šótˢm|	-V̄ Dynamic	**a* Actant	'to close' = action turning open → closed
CVCV́C	ϣⲧⲟⲙ |šˢtɔ́m|	+V̄ Non-dynamic	**a* Actant	'to be closed/closing' = ongoing state of being closed/closing
CVCV́C	ϣⲧⲁⲙ |šˢtám|	+V̄ Non-dynamic	**E* Non-actant/ External-agentive	'to have been closed/closing' = state caused by agent

Similarly, [S] ϩⲱⲛ |hón| < **ẖá(XV)n* / **ẖán(VX)* might have been originally a dynamic infinitive meaning 'to approach', while [Sa] ϩⲛⲁⲛ |hˢnán| < **ẖ(V)nā́n* could have been its static counterpart 'to be in an approaching state' = 'to be near'. This interpretation implies an [actant]/[no external agent] context (expressed by the vowel **a*) in both infinitives, and allows for both transitive and intransitive readings.

The verbs [S] ⲥⲛⲁⲧ |sˢnát| < **sVnÉḏ* : 'to fear', 'to be afraid' and [S] ϭⲛⲁⲧ |kʲˢnát| < **qVnÉḏ* 'to be angry' provide two additional good examples: 'to fear' and 'to be angry' can be understood as 'to be in the state of being afraid/angry because of something/someone', essentially a non-actant, non-dynamic infinitive/state verbal noun. The vowel **E* reflects that these emotional states are caused by an external agent, and can thus be interpreted as having low internal agency, which aligns well with the analysis of the vocalization proposed here.

23 The +V̄ infinitives of these verbs appear either in their free form or as part of [ⲛⲁ |na| + +V̄ infinitive] adjective verbs.

Non-dynamic interpretations might be suggested also for non-B-stems infinitives. Just to mention a few examples, the -V̄ infinitive A ⲥⲟⲟⲩⲉ |sɔ́ːhə| < **sáʿḥVʿ* could be understood as 'the act of causing to stand' or 'the act of setting up' (dynamic), while its +V̄ counterpart A ⲥⲉⲩⲟ |səhɔ́| < **sVʿḥáʿ* might perhaps represent 'the state of keeping standing' or 'the state of keeping upright' (non-dynamic). Similarly, the infinitive [A] ⲥⲟⲣϩⲩ |sᵊkɨréh(ᵊ)| < **sVgrÉ̂ḥVẏ* : 'to rest', 'to pause', 'to be quiet', 'to rest oneself', which appears to be a ẏ-extended (see below) s-stem of the pre-Coptic verb *grḥ* : 'to complete', 'to cease', 'to be satisfied', is clearly non-dynamic in meaning.

A quick look at the Coptic attestations of some of these forms shows that these interpretations may indeed fit a literal or etymological reading of the contexts in which they occur. For instance, for the series ϣⲱⲧⲙ |šótᵊm| / ϣⲧⲟⲙ |šᵊtɔ́m| / ϣⲧⲁⲙ |šᵊtám| :

Example 1 - ϣⲱⲧⲙ |šótᵊm| (pseudo.timothy > on_abbaton.02 : norm_group 532 - 542)[24]

"*And it was in this wise that My Father expelled Adam and Eve from Paradise and…*

ⲁ-	ϥ-	ϣⲱⲧⲙ	ⲙ-	ⲡ-	ⲣⲟ
|a-|	|f-|	|šótᵊm|	|ṃ-|	|pᵊ-|	|ró|
make-PST.AUX	3SG.M	act(ion)_of_closing [-V̄ + *a = dynamic action]	of/in/among[25]	the	door

Literal translation: '*…he made/performed the action of closing (of) the door…*'
Sense translation: '*…he closed the door…*'

… and He placed a mighty being of fire to watch the gate of Paradise, so that no one might enter therein."

‣ Comments: the focus here is clearly on the *act* of closing the door by God as part of a sequence of actions (1. 'expelled', 2. 'closed', 3. placed'), rather than on the result of the action itself.

Example 2 - ϣⲧⲟⲙ |šᵊtɔ́m| (sahidic.ot > 18_Job_38 : norm_group 60 - 70)

"*When the stars were made, all my angels praised me with a loud voice and…*

ⲁ-	ⲓ-	ϣⲧⲟⲙ	ⲇⲉ	ⲉⲣⲛ-	ⲑⲁⲗⲁⲥⲥⲁ	ϩⲛ-	ϩⲉⲛ-	ⲡⲩⲗⲏ
|a-|	|j-|	|šᵊtɔ́m|	|dé|	|ərn-|	|tʰálasːa|	|hṇ-|	|hən-|	|pýle|
make-PST.AUX	1SG	state_of_being_closed [+V̄ + *a = ongoing state]	then	for/against	sea	with/by	ART.INDEF.PL	door

Literal translation: '*…then I made/established a/the state of being closed off with doors for/against the sea…*'
Sense translation: '*…then I sealed off the sea with doors…*'

…when it rushed out, coming forth out its mother's womb."

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24 All the examples are taken from the Annis database https://corpus-tools.org/annis/ .

25 Synchronically, in Coptic, this preposition can be analysed as an object/accusative marker. Etymologically, however, it is likely derived from a partitive construction (Vycichl 1984: 134 with refs); hence the glossing 'of/in/among'.

‣ Comments: the sentence describes the establishment of a *state* of being closed that is affecting/directed toward the sea. The focus is not on the action itself, nor on its actor (not primarily, at least) but on the ongoing sealed state that had affected the sea since creation.

Example 3 - ⲁⲩⲧⲁⲙ |šˢtám| (apophthegmata.patrum > AP.038.macarius.04 : norm_group 24 - 34)

"And he answered, saying: 'I am Macarius.' And…

ⲁ-	ⲫ-	ⲁⲩⲧⲁⲙ	ⲙ-	ⲡ-	ⲣⲟ
|a-|	|f-|	|šˢtám|	|ṃ-|	|pˢ-|	|rɔ́|
make-PST.AUX	3SG	state_of_having_been_closed [+V̄ + *E = intrinsic state]	of/in/among	the	door

Literal translation: *'…he [Antony] made/caused the state of having been closed of the door…'*
Sense translation: *'…he [Antony] closed the door…'*

…and went in and left him."

‣ Comments: the sentence here can be understood as describing the door having become closed/closing as a *state* resulting from Antony's ('he') action.

It is clear that these observations are only preliminary and that a systematic study of such forms across a broader range of verbs would be required to clarify more precisely the semantic values encoded by these constructions.

5.4.3 Verbal Suffixes

Egyptian verbs also display two postponed elements: the suffixes *-ỷ* and *-(V)t*. Although both function as suffixes, the suffix *-ỷ* seems to have been present in various verbal forms (it is certainly recognizable both in infinitives and pseudoparticiples/statives in Coptic), while the suffix *-(V)t* appears to be strictly limited to infinitives (it is absent, for instance, in pseudoparticiples/statives). This suggest that while both these suffixes are appended to the verb, they might have been functioning at different linguistic levels, with *-ỷ* influencing the core meaning of the verb, while the *-(V)t* working at a more superficial verbal form level.

5.4.3.1 *The suffix -ỷ*

This suffix *-ỷ* is likely the most understudied verbal morpheme, and potentially the most controversial one. While scholars have long proposed its existence (see e.g. Lacau 1953), particularly in connection with 3-lit verbs, Egyptological grammars have generally ignored it altogether, classifying the verbs potentially displaying it as so-called final-weak verbs, i.e. taking the final glide as a root consonant.

The evidence supporting its existence, however, seems to me quite solid, and can be summarized as follow:

1. Some verbs that appear to have had three strong consonants and a final glide, and are therefore typically classified as so-called IV-weak verbs, are clearly denominal. Crucially, they appear to be derived from 3-lit nouns that originally showed no trace of a final glide in their root. Examples include *nṯrỷ* : 'to make/be divine' < *nṯr* : 'god' and *kȝrỷ* : 'to enshrine' < *kȝr* : 'shrine'.
2. The existence of a derivational verbal morpheme *-ỷ* is further supported by Coptic, where we find a few examples of verb pairs that clearly derive from the same root, but which are distinguished by vocalizations suggesting two distinct patterns: one characterized by only three consonants and another by three consonants plus a final glide. In some cases, slight differences in meaning have been noted between these verbs, but for the majority of them, dictionaries record no distinction in meaning between the two forms; e.g.: [B] ϭⲱⲙⲥ |tʰómˢs| < **ṯám(V)s* vs [B] ϭⲟⲙⲥ |tʰɔ́ms(ᵊ)| < **ṯámsVỷ* : both 'to bury'. Similar pairs are also attested among verbs that are traditionally classified as 2-lit , and which would be assumed to derive from 2⁺-EtR^{prtCpt} roots in the frame of this article; in these cases, the longer form displays a pattern **CáwCV*, with a extra *-w-* that appears as the second consonant, but is absent in other forms of the verb; e.g.: [S] ⲧⲱⲧ |tòt| < **tá(XV)t* or **tát(VX)* : 'to be joined, persuaded, agreeable', 'to conjoin, mingle' vs [S] ⲧⲟⲟⲩⲧⲉ |tɔwtə| < **tawtVỷ* : 'to be gathered, collected', 'to gather, collect'. As a side note, it is also worth noting that this extra *-w-* could perhaps be seen as further evidence for the existence of a third consonant *-X-* in these roots, since it appears precisely where such a hypothetical consonant might have been located within the root.

While the arguments supporting the existence of a suffix *-ỷ* seem to me quite clear, its exact meaning remains elusive. As mentioned, it appears to have a denominative function when associated with nominal roots. However, this is clearly not its only function, as *-ỷ* also appears as a suffix with clearly verbal roots. A systematic study of the semantics of verbs that appear to display this suffix would be necessary to better define its function, but such an analysis lies beyond the scope of this paper.

The suffix *-ỷ* is not limited to the infinitive but is also preserved in the pseudoparticiple/stative forms. In verbs with three strong consonants, it regularly appears after the root consonants. For instance, the verb *šms-ỷ* : '(to) worship', '(to) follow', has a Coptic pseudoparticiple/stative form [A2] ϣⲙϣⲏⲩ |šᵊmšéw| < **šVmsÉ-ỷ-V*. In contrast, in verbs with only two strong consonants, the suffix *-ỷ* appears to undergo metathesis with the final consonant of the root. This is exemplified, for instance, by the verb *tt* : '(to) join', '(to) gather', whose pseudoparticiple/stative has the form [S] ⲧⲟⲩⲏⲧ |tᵊwét(ᵊ)|. This form can be analysed as deriving from **tVwỷÉt-V* < ***tVwtÉỷ-V*. This vocalization can be explained assuming the medial consonant ⲟⲩ |w| was originally a geminate or a consonant cluster, which forced the stress to stay on the second syllable. Moreover, from a phonological perspective, one might suggest that the metathesis occurred because the *-w-* "attracted" the *ỷ* of the suffix.

Infinitives with the suffix *-ỷ* are attested with vocalizations in both **a* and **E*, both as +V̄ and -V̄. However, -V̄ infinitives with vocalization in **E* also exhibit an additional suffix *-(V)t* (see below for further discussion), whereas those with -V̄ vocalization in **a* as well as +V̄ ones do not. In Coptic, this *-(V)t* is typically lost in the absolute form of the

infinitive but is retained in the prepronominal form. For instance, with **E* and the suffix *-(V)t*:

šms-ỷ : '(to) worship', '(to) follow'
Absolute: [S] ϣⲙⲱⲉ |šm̀šə| < **šÉmsV-(ỷ)-t*
Prepronominal: [S] ϣⲙⲱⲛⲧⲣ |šᵊmšét≠| < **šVmsÉ-(ỷ)-t≠*

Versus with **a* and without the suffix *-(V)t*:

msq̲-ỷ : '(to) hate'
Absolute: [S] ⲙⲟⲥⲧⲉ |mɔ́stə| < **mástV-ỷ*
Prepronominal: [S] ⲙⲉⲥⲧⲱⲣ |məstóỷ≠| < **mVstá-ỷ≠*

It is important to note here that Coptic infinitives reflecting a pattern **CÉCCVt* may in fact derive from two distinct pre-Coptic patterns. This is because when the suffix *-(V)t* is present, the suffix *-ỷ* becomes effectively invisible in the corresponding Coptic infinitives, as it appears to be assimilated by the *-(V)t*.[26] This assimilation introduces a degree of ambiguity into the system, because, in principle, infinitives reflecting a form **CÉCCVt* could then derive either from a form with the *ỷ*-suffix **CÉCCVỷ(V)t*, or from a form **CÉCCVt* without it. The latter could be assumed to originate from a base form **CÉC(V)C* extended with the suffix *–(V)t*, where **CÉC(V)C* represents the E-vowel counterpart of the more common **CáC(V)C* pattern.

However, while identical in the absolute state of the infinitive, in principle **CÉCCVt* < **CÉCCVỷ(V)t* verbs can be expected to differ from real **CÉCCVt* ones in the pseudoparticiple/stative, as the first would display the suffix *-ỷ*, while the latter would show no trace of it. This is indeed precisely what we observe in the Coptic data.

In particular, on the one hand we have verbs like [S] ϣⲙⲱⲉ |šm̀šə| < **šÉmsV-(ỷ)-t* : 'to worship', which as mentioned above present a pseudoparticiple/stative that is compatible with the presence of the suffix *ỷ*, namely [A2] ϣⲙⲱⲛϒ |šᵊmšéw| < **šVmsÉ-ỷ-V*. On the other hand, we have verbs like [S] ⲥⲣⲫⲉ |sf́fə| < **sÉrf-Vt* : 'be at leisure', whose pseudoparticiple/stative [S] ⲥⲣⲟⲫⲧ |sᵊrɔ́ft(ᵊ)| < **sVráf-tV*[27] present no trace of a suffix *-ỷ*.

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26 The assimilation of expected final glides by *-(V)t* suffixes is a phenomenon that has been observed by scholars also in other contexts, although interpretations may vary. For instance, Loprieno 1995, 59 observes a similar phenomenon in the case of feminine nouns that display a pattern *CVCCV-t* in the singular, but *CVCCVỷ-wVt*, although he interpret them as nominal stems with final vowel, rather than as forms with a final ỷ that is assimilated to the *-(V)t* of the singular; these interpretations are functionally equivalent.

27 Note that while the stative [A2] ϣⲙⲱⲛϒ |šᵊmšéw| derives from pre-Coptic masculine pseudoparticiple, the form [S] ⲥⲣⲟⲫⲧ |sᵊrɔ́ft(ᵊ)| < **sVráf-tV* derives from a feminine one, the -ⲧ |-t(ᵊ)| < **-tV* being the feminine ending. The difference in gender is not problematic (it is well known that some verbs preserve a masculine form the pseudoparticiple, others a feminine one), and does not affect the argument, because if the suffix *-ỷ* were present, the feminine pseudoparticiple should have been *□sVrfV́-ỷ-tV*, not **sVráf-tV*, as attested by e.g. [S] ⲟⲁⲗⲏⲩⲧ |kʲaléwt(ᵊ)| < **qVỷrÉ-ỷ-tV*, the feminine pseudoparticiple of the verb [S] ⲟⲩⲉⲓⲗⲉ |kʲɔ́jlə| < **qáỷrV-ỷ* : 'to dwell'.

From this evidence, one can thus infer that while [A2] ⲥⲙⲱⲛϒ |šᵊmšéw| derives from a form **CÉCCVt* < **CÉCCVy̆(V)t*, originally with *-y̆*, [S] ⲥⲣⲫⲉ |sṛ́fə| must instead reflect a form that was originally **CÉCCVt* and never had *-y̆* to begin with.

All this considered, it becomes now possible to organize all these forms into a coherent and symmetrical paradigm, as in table 4.

Table 4 | **a* / **E + Vt* opposition in 3-EtR^{prtCpt} verbs.

Pattern	**a*	**E + Vt*
CVC(V)C	**CáC(V)C* E.g.: [B] ⲑⲱⲛⲥ |tʰómᵊs| < **t̲ám(V)s*	**CÉCC-Vt* E.g.: [S] ⲥⲣⲫⲉ |sṛ́fə| < **sÉrf-Vt*
CVCCVy̆	**CáCCVy̆* E.g.: [B] ⲑⲟⲛⲥ |tʰóms(ᵊ)| < **t̲ámsVy̆*	**CÉCC-Vt* < *CÉCCVy̆-(V)t* E.g.: [S] ⲥⲙⲱⲉ |šḿ̩šə| < **šÉmsV-(y̆)-t*

Infinitives that can be analysed as *y̆*-extended exist also as variants of +V̄ forms, but they are exceedingly rarer. One can mention, for instance, [S] ϭⲣⲱϩ |kʲᵊróh(ᵊ)| < **gVráḥ-Vy̆* : ‘to be in want, needy’ and [S] ϩⲣⲏϭ |hᵊrékʲ(ᵊ)| < **gVrÉḥ-Vy̆* = ‘to be still’, ‘to cease’ for 3-EtR^{prtCpt} verbs, and [B] ϩⲣⲟⲩⲣ |hᵊrúr(ᵊ)| < **hVrár-Vy̆* : ‘to cease’, ‘to be quiet’ and [S] ⲧⲉⲗⲏⲗ |təlél(ᵊ)| < **t̲VrÉr-Vy̆* : ‘to rejoice’ for 2⁺-EtR^{prtCpt} verbs, assuming as argued above that so-called geminated verbs are only a paradigmatic manifest of primarily two-consonant roots.

Interestingly, positing the existence of an extended *-y̆* form offers an elegant solution to a longstanding issue in Egyptian historical morphology and phonology (see e.g. Lacau 1953; Osing 1976, 51–54; Allen 2020, 127). This issue concerns the infinitives of so-called III-weak verbs, which appear in Coptic in two distinct forms. One form exhibits a vocalization suggesting that the verb had three root consonants, the last of which being a weak consonant (e.g. [S] ⲡⲣⲣⲉ |pṛ́rə| < **pEry̆-Vt* : ‘to come out’, ‘to move, put out’, ‘to blossom (of plants)’), while the other has only two consonants, with no trace of a final weak one (e.g. [S] ⲡⲓⲣⲉ |pírə| < **pEr-Vt*[28] : ‘to come out’, ‘to move, put out’, ‘to blossom (of plants)’). Notably, the forms lacking the final weak consonant are far more common, whereas those with the weak consonant are attested only in a small subset of verbs.

Clearly this issue is directly tied to the classification of these verbs as III-weak—i.e., as verbs with three root consonants, the last being a weak consonant. If this classification were accurate, why do the most common infinitive forms show no trace of a final weak consonant? At the same time, if the loss of the final weak consonant were a regular development, why are there also forms that retain it, often attested for the same verb and within the same Coptic dialects that also exhibit the shorter forms without it?

However, if we consider that, as mentioned, the traditional definition of verbal classes is highly questionable, and more specifically, if we interpret verbs traditionally

28 Note that both these types of infinitives had an additional suffix *-(V)t*; the suffix *-(V)t* was lost in the absolute form of the infinitive, but in most dialects, it was preserved in the prepronominal and prenominal form. This suffix *-(V)t* will be discussed here below.

classified as IV-weak as *ỷ*-extended forms of what are actually 3-EtR roots, a simple solution to these problems emerges, namely the same solution already emerged above for the so-called geminated verbs: one may suggest that so-called III-weak verbs are an Egyptological mirage and simply do not exist as a distinct verbal class, and they should instead be understood as verbs derived from 2⁺-EtR^{prtCpt} roots[29] that can appear both in the basic stem **CÉ(X)C-Vt* or **CÉC(X)-Vt* (the most common forms in Coptic, without an extra final *-ỷ*) and, more rarely, in a *ỷ*-extended from **CÉ(X)C-ỷ-Vt* or **CÉC(X)-ỷ-Vt* (the rarer forms in Coptic, with an extra final *-ỷ*). Both these forms are characterized by an additional suffix *–(V)t* (see below).

If the analysis suggested here were correct, and so-called III-weak verbs were truly one and the same class as so-called 2-lit verbs, we would expect to find at least some of these verbs displaying forms typical of so-called 2-lit verbs. Indeed, this is what can be observed in Coptic, where a few verbs usually classified as so-called III-weak display both the infinitives just discussed and an infinitive analogous to that typical of so-called 2-lit verbs. For instance:

[S] ϣικε |šíkə| < **šÉ(X)q-Vt* or **šÉq(X)-Vt* : ‘to dig’
[S] ϣωκ |šók| < **šá(XV)q* or **šáq(VX)* : ‘to dig’, ‘to be deep dug’

[A] ϯκε |tíkə| < **tÉ(X)k-Vt* or **tÉk(X)-Vt* : ‘to throw’
[A] ϯωκ |tók| < **tá(XV)k* or **ták(VX)* : ‘to throw’

[S] ϣιτε |šitə| < **šÉ(X)d-Vt* or **šÉd(X)-Vt* : ‘to demand’, ‘to extort’
[S] ϣωτ |šót| < **šá(XV)d* or **šád(VX)* : ‘to demand’, ‘to extort’

Similarly, there are at least two[30] cases of verbs that display both an infinitive *CÉ(X)C-Vt* or *CÉC(X)-Vt* and an infinitive with a pattern **CáwCV*[31] analogous to that of some so-called 2-lit verbs. The first is the verb [A] ⲡⲁⲩⲣⲉ |páwrə| < **pawrVỷ* : ‘to increase’, ‘to grow (of animals and plants)’, likely related with [S] ⲡⲓⲣⲉ |pírə| < **pir-Vt* : ‘to come out’, ‘to move, put out’, ‘to blossom (of plants)’. The second is the descendant of Egyptian *ḏỷ* < *ḏd*[32] : ‘(to) say’, ‘(to) speak’, which present three Coptic infinitives. The first two are

29 This analysis may also lend further support to the idea that 2⁺-EtR^{prtCpt} roots were in fact *CXC* roots, with *-X-* being some kind of weak consonant. In particular, Vycichl (1952: 1, n2) noted that a few Egyptian verbs traditionally classified as III-weak correspond to *C-w-C* roots in semitic (e.g., Eg. *ʿpj* ~ Sem *ʕwp* : ‘(to) fly’, Eg. *ṯnj* ~ Sem *kwn* : ‘(to) rise’, Eg. *ỉbj* ~ Sem *lwb* : ‘(to) be thirsty’). He did not provide an explanation for the apparent misplacement of the weak consonant. Under the model suggested here, however, no such explanation would be necessary, because the Egyptian forms would not have a final weak consonant but rather a medial one, just like their Semitic counterparts, being *ʿXp*, *ṯXn*, and *ỉXb* respectively.

30 A third candidate is likely [A2] ϫⲁⲩⲗⲉ |čáwlə| < **ṯáwnVỷ* : ‘to support’. Its pre-Coptic ancestor, *ṯn* : ‘to lift up’, is traditionally interpreted as a III-weak verb, as it is the case for the verb *pr* : ‘to come/move out’. However, [A2] ϫⲁⲩⲗⲉ |čáwlə| seems to be the only infinitive of this verb surviving in Coptic, and therefore this verb is of little help for the present discussion, as it provide no direct evidence for the coexistence of different infinitives for the same verb.

31 Here as well, with *-w-* < *-X-* ?

32 The pre-Coptic Egyptian verb is usually assumed to have been *ḏd*, as its hieroglyphic spelling is  . However, most of its Coptic descendants point to a protoform **ḏỷ* without final *-d*. The

1. [S] ⲭⲱ |čó| < *ḏá(ⲬV)ẏ́ or *ḏáẏ́(VⲬ) ; 2. [A2] ⲭⲟⲟⲩⲉ |čɔ̀wə| < *ḏáwẏ́-Vẏ́. This verb has also a third infinitive, which however is not attested in the absolute state, but only in the prenominal and prepronominal states. Its forms are:

Absolute	Prenominal	Prepronominal
—	[S, A, F] ⲭⲓ- |čɨ-| or |čʾj-| [S, A, sA, B, F] ⲭⲉ- |čə-| [F] ⲭⲁ- |ča-|	[S] ⲭⲓⲧ= |čít=|

This paradigm appears to correspond to that attested for the pattern *C¹EC²-Vt* for verbs that are either etymologically *C-ẏ́/w/ʒ́*, such as:

Absolute	Prenominal	Prepronominal
[S etc.][33] ⲩⲓ |ší| < *ẖÉʒ́-Vt : 'to measure, weigh', √ẖʒ́[prtEg]	[S etc.] ⲱⲓ- |ši-| or |šʾj-|	[S etc.] ⲱⲓⲧ= |šít=|
[S etc.][34] ϩⲓ |hí| < *ẖÉẏ́/w-Vt : 'to thresh, beat, rub', √ẖẏ́/w[prtEg]	(□ⲉⲓ- |hɨ-| or |hʾj-| ?)[35]	[S etc.] ϩⲓⲧ= |hít=|

Or *C-ẏ́/ʒ́*[36] < *i̯-C*,[37] such as:

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development *-d#* > *-ẏ#* is irregular, as *-d* is usually preserved in final position, but it could have been influenced by the development *-t#* > *-ẏ#* , which is instead regular. As for when this change took place, we know that 𓂝 was read as *ḏVẏ́* already in the Mid/Late New Kingdom, as this is the value it has in Group Writing; see e.g. the spelling 𓂝𓏏 = *ḏẏ́-t* > [S] ⲭⲟⲉⲓⲧ |čójt| : 'olive', where the group 𓂝 is transcribing the segment **ḏaj-* of the word **ḏajtV* > [S] ⲭⲟⲉⲓⲧ |čójt| (see also Hoch 1994 with additional examples, although he reads this group as *ḏi* rather than *ḏVẏ́*). However, the phonetic change itself is likely earlier, as there are sporadic spellings of the verb as just 𓌃, without *-d*, already in the Late Second Intermediate Period/Early New Kingdom (e.g. the stele of Emhab [JE 49566] from Edfu, dated to the reign of Kamose – see the TLA, Objekt-ID SGACEU3735FONLTWNXWREEXX2E). At the same time, it is worth pointing out that the *-d#* seems to be preserved as ⲧ |t| in some forms in some Coptic dialects (participle: [S, A2] ⲭⲁⲧ- |čat-|, possibly prepronominal imperative [B] ⲁⲭⲟⲧ= |ačɔ́t=| - note that there is also a form [B] ⲁⲭⲟ= |ačɔ́=|, and all other dialects have imperative forms without ⲧ |t|). These instances might be due to dialect-specific archaisms or to specific phonological environments that cause the final *-d#* to be preserved in these forms.

33 This verb is also attested in other dialects, but I cite only the Sahidic forms here for the sake of simplicity.

34 This verb is also attested in other dialects, but I cite only the Sahidic forms here for the sake of simplicity.

35 I mark this form with □ because it is the expected but not attested form: Crum's dictionary (1939: 643a) list no prenominal form for this verb. However, a prenominal form ϩⲓ- |hi-| / |hʾj-| is attested in S, B, and F, but it is usually interpreted as a form of the related and semantically similar verb [S] ϩⲓⲟⲩⲉ |híwə| : 'to strike', 'to be struck', for which a prenominal form [S, A, A2] ϩⲟⲩ- |hu-| / |hʾw-| is also attested. One may wonder: could it be that ϩⲓ- |hi-| / |hʾj-| was the prenominal form of ϩⲓ |hí|, and only ϩⲟⲩ- |hu-| / |hʾw-| truly belongs to ϩⲓⲟⲩⲉ |híwə|?

36 Unless the use of *ʒ́* is just a Late Egyptian orthographic phenomenon, representing the glide I transcribe as *ẏ́* in this paper.

37 To my knowledge, this phonological development has not been noted before; it is currently impossible to say when it took place and if it should be seen as dialect-specific development.

Absolute	Prenominal	Prepronominal
[S etc.][38] ⲭⲓ |čí| < *<u>ṯ</u>*É<u>ỷ</u>*-Vt* : ‘to take’, ⳕ*<u>ṯ</u>ỷ*^{LE} < √*<u>ỉṯ</u>*^{prtEg}	[S etc.] ⲭⲓ- |či-| or |čᵊj-|	[S] ⲭⲓⲧ= |čít=|
[F][39] ⲛⲓ |ní| < *n*É*ỷ-Vt* : ‘to take’, ⳕ*nỷ*^{prtCpt} < √*ỉn*^{prtEg}	□ⲛⲓ- |ni-| or |nᵊj-|[40]	[F] ⲛⲓⲧ= |nít=|

Based on these parallels, it is reasonable to assume that the absolute form of the third infinitive of the verb *<u>ḏ</u>ỷ* < *<u>ḏ</u>d* might have been [S] □ⲭⲓ \|čí\| < **<u>ḏ</u>*É*ỷ-Vt*.[41]

Remarkably, the root √*pr* ‘(to) come/move out’ is traditionally classified as III-weak, while the root √*<u>ḏ</u>ỷ* < √*<u>ḏ</u>d* is traditionally classified as 2-lit . This is a prime example of how inconsistent and erratic the traditional classification of verbal classes can be. These data can be summarized as in table 5.

Table 5 | Paradigm for 2⁺-EtR^{prtCpt} verbs in the **Cá(XV)C* / **CáC(VX)*, **CáwC-Vỷ*, **CÉ(X)C-Vt* / **CÉC(X)-Vt*, **CÉ(X)C-ỷ-Vt* / **CÉC(X)-ỷ-Vt* forms.

2⁺-EtR^{prtCpt}	**Cá(XV)C* or **CáC(VX)*	**CáwC-Vỷ*	**CÉ(X)C-Vt* or **CÉC(X)-Vt*	**CÉ(X)C-ỷ-Vt* or **CÉC(X)-ỷ-Vt*
šq	[S] ⲱⲱⲕ |šók| < **šá(XV)q* or < **šáq(VX)*		[S] ⲱⲓⲕⲓ |šíkɨ| < **šÉ(X)q-Vt* or < **šÉq(X)-Vt*	
tt	[S] ⲧⲱⲧ |tót| < **tá(XV)t* or < **tát(VX)*	[S] ⲧⲟⲟⲩⲧⲉ |tówtə| < **táwt-Vỷ*		
pr		[A] ⲡⲁⲩⲣⲉ |páwrə| < **páwr-Vỷ*	[S] ⲡⲓⲣⲉ |pírə| < **pÉ(X)r-Vt* or < **pÉr(X)-Vt*	[S] ⲡⲣⲣⲉ |pṛ́rə| < **pÉ(X)r-ỷ-Vt* or < **pÉr(X)-ỷ-Vt*

But if Late Egyptian *<u>ṯ</u>ỷ* : ‘to take’ is truly derived from the same root as Middle Egyptian *<u>ỉṯ</u>* : ‘to take’ (as I am inclined to believe), then the change must have taken place before its earliest New Kingdom/Late Egyptian attestations, at least in the dialect underlying those texts. However, these observations are rather preliminary, and further studies would be needed.

38 This verb is also attested in other dialects, but I cite only the Sahidic forms here for the sake of simplicity.

39 These forms are attested only in Fayyumic, alongside the more common [F] ⲓⲛⲓ \|íni\| < **iÉn-Vt*. The latter is the only form attested in other dialects, namely [S, A] ⲉⲓⲛⲉ \|íne\|, [A, A2] ⲓⲛⲉ \|íne\|, [B, F] ⲓⲛⲓ \|íni\|.

40 Only ⲉⲛ- \|ən-\| is found in Fayyumic; this is more plausibly derived from [F] ⲓⲛⲓ \|íni\| than from [F] ⲛⲓ \|ní\|.

41 Some scholars (e.g. Vycichl 1984, 323–24) reconstruct the absolute infinitive of this verb as **<u>ḏ</u>id* (= **<u>ḏ</u>É<u>d</u>* in my system) rather than **<u>ḏ</u>É<u>ỷ</u>-Vt*, likely interpreting the final ⲧ \|t\| of the prepronominal form as deriving from the original final *-d*. I have no formal argument to reject this hypothesis. I can only say that there is no unambiguous evidence that such a vocalic pattern *CÉC* truly existed for any other verbs; the main other example, namely [A, A2] ⲟⲩⲉⲛ \|wén\| as alternative infinitive of the verb [S, etc] ⲟⲩⲱⲛ \|wón\| : ‘to open’ was questioned already by Sethe (1929: 69, 71). By contrast, the pattern **CÉỷ/w/ɜ-Vt* is well supported by the provided parallels. This, in my opinion, makes a reconstruction **<u>ḏ</u>É<u>ỷ</u>-Vt* preferable.

ḏ̱ỷ < *ḏ̱d*	[S] ꜣꜥ |čó| < **ḏ̱á(ⲬV)ỷ** or < **ḏ̱áỷ(VⲬ)**	[A2] ꜣꜥooγє |čɔ́wə| < **ḏ̱áwỷ-Vỷ**	[S] ⸢ꜣ|čí| < **ḏ̱É(Ⲭ)ỷ-Vt** or < **ḏ̱Éỷ(Ⲭ)-Vt**

It is interesting to observe that in these examples, the forms that can be reconstructed with the vowel **E* in the first syllable also display the suffix *-(V)t*, while those that can be reconstructed with a vowel **a* do not display it. This distribution is hardly accidental, and it will be discussed further below in connection with the *-(V)t* suffix.

The data in table 5 and their organization suggest another crucial insight: if we ignore for a moment the suffix *-(V)t*, the forms **CÉ(Ⲭ)C(-Vt)* / **CÉC(Ⲭ)(-Vt)* and **CÉ(Ⲭ)C-ỷ(-Vt)* / **CÉC(Ⲭ)-ỷ(-Vt)* appear as obvious **E* counterparts of the forms in **a* **Cá(ⲬV)C* / **CáC(VⲬ)* and **CáwC-Vỷ*, thus resulting, once again, in a coherent and symmetrical paradigm.

Finally, while the active nature of the forms with **a* is clear and expected, the semantics of the forms with **E* is less straightforward. Based on the discussion above, one might expect these **E* infinitives to convey passive or non-agentive meanings, but this does not seem to be the case. At the same time, as noted, these forms also uniformly exhibit the suffix *-(V)t*, and it is reasonable to assume that this suffix had some influence on their meaning.

It is therefore essential now to better define the characteristics of this suffix *-(V)t*.

5.4.3.2 *The suffix -(V)t*

The suffix *-(V)t* appears only on the infinitives of some verbs. First of all, it is important to stress that even though it has often been interpreted as such, this is not a feminine suffix *-t*; this is clear from that fact that infinitives displaying it are regularly behaving as masculine nouns wherever their gender is evident,[42] never as feminine (see e.g. Gardiner 1957: 223).

The suffix *-(V)t* has been traditionally interpreted as a marker of infinitives typical of some verbal classes (especially classes with supposed weak final consonants). This analysis is problematic both within the traditional framework and in the approach proposed here. First of all, as discussed earlier, verbs traditionally classified as III-weak and IV-weak rarely show their supposed weak consonant clearly spelled out. Consequently, the suffix *-(V)t* is often treated as a diagnostic feature of such final-weak verbs, even when there is no other evidence for the presence of such final consonant. This reasoning is clearly circular and thus inherently flawed.

Moreover, as noted above (§5.4.3.1.), at least some of these verbs have infinitive forms that do not feature the suffix *-(V)t* and are indistinguishable from the infinitives of verbs traditionally classified as 2-lit. This further undermines the notion that *-(V)t* can function as a reliable verbal class marker. Such an interpretation would have been plausible if

42 For instance, they are specified by the masculine article in Late Egyptian, not the feminine one. There are a few rare cases of Old/Middle Egyptian forms that superficially look like infinitives in *-(V)t* that seem to behave as feminine, but as already Gardiner pointed out, these may not be truly infinitives to begin with (see e.g. Gardiner 1957: 223).

Egyptian-Coptic verbs had only one infinitive form; however, the evidence clearly shows that a single verb can exhibit multiple infinitives, some with *-(V)t* and others without it. Therefore, since the relationship between a given verb and the presence or absence of *-(V)t* in its infinitives is not exclusive, the presence or absence of *-(V)t* itself cannot be taken as a reliable verbal class indicator.

Things are even less convincing if the framework presented here is taken into account: if the so-called III-weak and IV-weak verbs are not actual classes but rather manifestations of 2⁺-EtR^{prtCpt} and 3-EtR^{prtCpt} verbs with or without the suffix *-ỷ*, then the very notion that the suffix *-(V)t* functions as a "verbal class marker" falls apart.

In turn, if this suffix is not a verbal marker, it is reasonable to assume that it carries some semantic or grammatical value. In this respect, a few observations are relevant. First, while the suffix *-(V)t* may not mark a verbal class, it is clearly associated with vocalizations in **E*. This correlation can be observed throughout the verbal system:

- 2⁺-EtR^{prtCpt} verbs, e.g.: [S] ⲡⲓⲣⲉ |pírə| < **pÉr-Vt*[43] : 'to come out', 'to move, put out', 'to blossom (of plants)'),
- *ỷ*-extended 2⁺-EtR^{prtCpt} verbs, e.g.: [S] ⲡⲣⲣⲉ |pṛ́rə| < **pÉrỷ-Vt* : 'to come out', 'to move, put out'
- 3-EtR^{prtCpt} verbs, e.g.: [B] ϩⲉⲙⲥⲓ |hémsi| < **ḥÉms-Vt* : 'to sit'
- *ỷ*-extended 3-EtR^{prtCpt} verbs, e.g.: [S] ⲱⲙϣⲉ |šémsə| < **šEmsV-ỷ-t* : 'to serve', 'to worship'
- *ỷ*-extended mediopassive causatives,[44] e.g.: [S] ⲥⲁϩⲛⲉ |sáhnə| < **sEḥnVỷ-t* : 'to cause oneself to take care'

At the same time, it is important to note that all forms displaying the suffix *-(V)t* consistently show stress patterns that agree with the standard syllabic rules and which can therefore be analysed as -V̄. In contrast, there is absolutely no evidence for the presence of the suffix *-(V)t* in any +V̄ verbal form. This distribution seems to hold even in cases where both types of infinitives are attested for the same verb. For instance, the three infinitives of the verbal root √*ḥms* : '(to) sit'[45] can be reconstructed as:

1. [B] ϩⲉⲙⲥⲓ |hémsi| < **ḥEms-Vt* — with stressed **E* in the first syllable (-V̄), and with *-(V)t*
2. [A] ϩⲙⲉⲥ |ḥˢmés| < **ḥVmÉ́s* — with stressed **E* in the last syllable (+V̄), and without *-(V)t*
3. [F] ϩⲙⲁⲥ |ḥˢmás| < **ḥVmá́s* — with stressed **a* in the last syllable (+V̄), and without *-(V)t*

The same can be observed also with 2⁺-EtR^{prtCpt} verbs (traditionally III-weak), as illustrated, for instance, by the verbal root √*ỉr* : 'to do':

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43 Note that both these types of infinitives had an additional suffix *-(V)t*; the suffix *-(V)t* was lost in the absolute form of the infinitive, but in most dialects, it was preserved in the prepronominal and prenominal form. This suffix *-(V)t* will be discussed here below.

44 See Kilani (forthcoming a).

45 For a discussion of the forms of this verb, see Peust (2022).

1. [F] ⲗⲁ |lá| < **íVrár*[46] — with stressed **a* in the last syllable (+V̄), and without *-(V)t*; this infinitive appears to have add an habitual/frequentative meaning (Vycichl 1984, 169)
2. [S] ⲉⲓⲣⲉ |íɾə| < **íÉr-Vt* — with stressed **E* in the first syllable (-V̄), and with *-(V)t*; this is the most common infinitive in Coptic

This further indicates that we are dealing with different verbal forms within a single verbal class, rather than with distinct verbal classes. It also reinforces the idea that the presence or absence of a final *-(V)t* is not a reliable diagnostic for verbal classification, as a single verb can exhibit both forms with and without it.

As for the possible function and meaning of this suffix *-(V)t*, it has been recently observed (Kilani forthcoming b) that verbs traditionally classified as III-weak, which can thus be assumed to have at least one infinitive with the suffix *-(V)t*, are neither uniformly nor randomly distributed across semantic fields. Rather, they tend to cluster in semantic fields that, in other Afroasiatic languages, are often associated with mediopassive, reflexive, or autobenefactive meanings. This may, of course, reflect an indirect correlation, stemming from the association of the vowel **E* with these semantic nuances (as discussed above), and the clear link between the suffix *-(V)t* and vocalizations involving **E*. However, it is intriguing to observe that in many Afro-Asiatic languages, mediopassive, reflexive, and autobenefactive nuances are frequently marked through a common morpheme **t*, which can likely be traced back to Proto-Afroasiatic. In fact, Egyptian itself has a morpheme *t(ỷ)* that appears to function either as a passive marker or as an indefinite agent pronoun/marker, and which has been suggested to be somehow connected with the mediopassive/autobenefactive Afro-Asiatic morpheme **t* (see e.g. Stauder 2023).

It is also important to note that this passive/indefinite agent marker *t(ỷ)* appears only with verbal forms that are usually interpreted as not being infinitives. In other words, the passive/indefinite agent marker *t(ỷ)* seems to be in complementary distribution with the infinitive suffix *-(V)t*, as there are no known instances in which both *t(ỷ)* and *-(V)t* occur together on the same verbal form. This distribution is suggestive and raises an intriguing possibility: perhaps the passive or indefinite agent marker *t(ỷ)* and the infinitive suffix *-(V)t* are not distinct morphemes, but rather two manifestations of the same single morpheme.

All these considered, one may wonder if, in Egyptian, the vocalization in **E* and the ending *-(V)t* served complementary roles: the former marking that the main argument of the verb is a non-actant experiencer (gloss: NONACT) of an action initiated by an external agent, and the latter indicating that this external agent is either undefined/unspecified (gloss: INDF/IMPERS) or the main argument itself (i.e., reflexive – gloss: REFL).

This analysis would explain why infinitives with a nonactant vowel **E* seem to have an active meaning. In particular, within this interpretative framework, an infinitive such as [S] ⲙⲁⲥⲧϥ |mástˤf| < *ms-t≠f : *mÉs-t≠Vf*, typically analysed as [birth-INF≠3SG.M] and translated as 'to birth him' or 'the birthing of him', could instead be understood as **m⟨É⟩s-t≠Vf* = [birth.INF.⟨NONACT⟩-INDF/IMPERS≠3SG.M], literally meaning 'his being born by someone' or 'to have him born by someone'. Similarly, a form like *pr-t≠f : *pÉr-t≠Vf*, typically analysed as [go_forth-INF≠3SG.M] and translated as 'to go forth of/by him' or 'his act of going forth' could

46 The loss of final *-r* in posttonic position and the *r* ~ *l* correspondence in Fayyumic are regular.

instead be analysed as **p⟨É⟩r-t≻Vf* = [moving_forth.INF.⟨NONACT⟩-REFL≻3SG.M], literally meaning 'his being moved forth by himself' or 'to move himself forth'.

5.4.4 Morphological Elements: Summary

As shown in the preceding analysis, the system of pre-Coptic infinitives appears to have been characterized by the following morphological oppositions and elements:

- Vocalic opposition -V̄ / +V̄
- Vocalic opposition **a/*E*
- Suffix *-ỷ*
- Suffix *-(V)t* (only with some infinitives)

Each of these elements seems to have carried distinct morphosemantic functions, though some are more clearly understood than others. When combined together, they form a coherent and symmetrical paradigm for Coptic infinitives, as illustrated in table 6.

Table 6 | Morphological paradigm for reconstructed for pre-Coptic infinitives.

-V̄			
**a*		**E + (V)t*	
-ỷ	*+ỷ*	*-ỷ*	*+ỷ*
CáC(V)C	CáCC-Vỷ	CÉCC-Vt	CÉCC-Vỷ-Vt

+V̄			
**a*		**E*	
-ỷ	*+ỷ*	*-ỷ*	*+ỷ*
CVCáC	CVCáC-Vỷ	CVCÉC	CVCÉC-Vỷ

It is important to emphasize once again that this analysis and the paradigms presented here have been constructed primarily on the basis of infinitives/verbal nouns, and therefore apply chiefly to those verbal forms. Other morphemes, and possibly additional oppositions, may have existed in other verbal forms. Investigating those aspects of the verbal system remains a task for future research.

6 Defining New Verbal Classes

As mentioned at the outset, this paper had a twofold aim: first, to provide a new framework for analysing the Coptic and pre-Coptic verbal system; second, to propose a redefinition of verbal classes. The answer to this second objective has, in fact, already emerged in the preceding sections. In particular, the early part of this paper has argued the following points:

- Verbs exhibiting *s-* and *n*-prefixes should be treated as derived stems of underlying verbs, rather than as members of entirely separate verbal classes.

- Similarly, reduplicated verbs represent morphological variants of the underlying roots, not independent verbal classes.
- The traditional class of so-called IV-weak verbs does not exist; rather, the relevant verbs are more accurately understood as 3-EtR^{prtCpt} verbs extended with the suffix -*ỷ*.
- The class of so-called III-weak verbs is likewise not a valid category; the verbs traditionally assigned to it are better analysed as 2^{+}-EtR^{prtCpt} verbs, which can be occasionally extended with the suffix -*ỷ*.
- The so-called II-geminate verb class is also not a genuine class, as the verbs grouped under it can be accounted for as one of the possible paradigmatically-conditioned surface shapes that 2^{+}-EtR^{prtCpt} verbs may exhibit in specific verbal forms.

When considered together, these observations suggest that the model presented here accounts for all the infinitives traditionally assigned to five or more distinct verbal classes[47] by positing only two core classes: one consisting of pre-Coptic roots with two strong consonants and possibly a third weak consonant (labelled here 2^{+}-EtR^{prtCpt}) and one of roots with three consonants (3-EtR^{prtCpt}).

Table 7 illustrates how these two classes combine with the morphological paradigm of the infinitives summarized in §5.3.4 to produce a complete paradigm for each class. As shown, not only are all the forms predicted by this model attested for some Coptic verb, but many of them also occur with the same verbs. This strongly supports the conclusion that these forms represent paradigmatic variations within a single verbal class, rather than distinct manifestations of separate verbal classes.

As pointed out at the beginning, non-reduplicated verbs with more than three strong consonants have not been discussed here, as Coptic provides very limited evidence for them. However, by analogy, it is reasonable to assume that such verbs formed a single four-strong-consonant class, with *ỷ*-extended or geminated forms functioning as regular morphological variants within that class. However, a systematic reassessment of the evidence would be needed to confirm this assumption.

At the same time, since the morphological elements and paradigms outlined above can be reconstructed from Coptic, yet were no longer productive at that stage, they must have been active in an earlier, pre-Coptic phase of the language. It is therefore reasonable to assume that they go back at least to the historically attested stages of Ancient Egyptian. However, it is difficult to determine when exactly the whole system was still productive and when it began to become lexicalized. As discussed at the beginning of this paper, it is likely that some features of the system remained functional and productive while others had already become fossilized (see, e.g. the opposition between *n*-stems and *s*-stems, with the latter possibly surviving as a productive category longer than the former). Yet, the rarity of verbs preserving more than one infinitive, combined with the scarcity of evidence for multiple distinct infinitives in Middle and Old Egyptian, strongly suggests that the

47 Namely 3-lit , IV-weak, 2-lit , III-weak, II-gem — or more if *s*- and n-prefixed and reduplicated verbs are treated as separate classes.

original system must have been quite ancient and perhaps even prehistoric. Interestingly, this situation is not unique. For instance, it finds a very good parallel in Ethiopic Semitic, where the historically rich system of Semitic stems is no longer productive. There, most verbs preserve only a single stem that has been lexicalized into new paradigmatic classes, which retain little of their original morphosemantic values.[48]

These considerations also impact the chronological definition of verbal classes, since, as noted above, verbal classes are a dynamic reality and have certainly evolved over time rather than constituting a fixed or absolute system. This means that although the types and number of verbal classes implied by a given morphosemantic model likely reflect those that were functionally present when the model itself was productive, the verbal system itself may yet have differed in earlier, more ancient stages of the language.

For instance, one might wonder whether the roots with 2^{+}-EtR^{prtCpt} and 3-EtR^{prtCpt} were truly the starting point of the system, or if something else can be reconstructed for even earlier phases. While Egyptian and Coptic may still preserve some clues to explore these questions,[49] the best way to approach them would be through comparison with other Afroasiatic languages. This, however, is clearly beyond the scope of the present paper and perhaps even beyond the current state of the field, and must therefore be left to future research.

Last but not least, it is clear that the arguments presented in this article have major consequences also for our interpretation and understanding of verbal morphology. Presumed paradigmatic alternations among verb belonging to supposedly different traditional verbal classes is often a primary means to define and distinguish most verbal forms (participles, so-called *sḏm=f* forms, etc.) in Old and Middle Egyptian. However, if the definition of verbal classes is questionable, then also the verbal forms based on them are unlikely to be accurate, and if the system of verbal classes is revised, then our whole understanding of the verbal morphology would need to be reassessed.

48 In some respects, these Ethio-semitic paradigmatic classes resemble the traditional Egyptological 'verbal classes': a primarily synchronic system built upon fossilized remnants of an earlier, richer system of roots, stems, prefixes, and suffixes.

49 For instance, as said, one may wonder if the infixed *-w-* appearing in the **CáCC-Vẏ* pattern of 2^{+}-EtR^{prtCpt} is a secondary element or a remnant of an earlier stage in which these roots were in fact three-consonant roots with a weak middle consonant. If this were truly the case, this development could be represented as 2^{+}-EtR^{prtCpt} < 3-$EtR^{pr\text{-}prtCpt}$.

BASE:	*CáC(V)C*	*CáCC-Vẏ*	*CÉCC-Vt*	*CÉCC-Vẏ-Vt*	*CVCā́C*	*CVCā́C-Vẏ*	*CVCḗC*	*CVCḗC-Vẏ*
3-EtR prtCpt	*CáC(V)C*	*CáCC-Vẏ*	*CÉCC-Vt*	*CÉCC-V(ẏ)-t*	*CVCā́C*	*CVCā́C-Vẏ*	*CVCḗC*	*CVCḗC-Vẏ*
gr̲ḥ	[S] ꞓɷpᴋ |hórᵊk| < **gár(V)ḥ*	[Sf] ꞓapᴋ |hárk(ᵊ)| < **gárḥ-Vẏ*			[S] ᴋpoᵹ |kᵊróh| < **gVrā́ḥ*	[S] ɔpɷᵹ |kʲróh(ᵊ)| < **gVrā́ḥ-Vẏ*	[S] ꞓpaᴋ |hᵊrák| < **gVrḗḥ*	[S] ꞓpнɔ |hᵊrékʲ(ᵊ)| < **gVrḗḥ-Vẏ*
ḫtm	[S] ɷɷтн |šótᵊm| < **ḫát(V)m*				[S] ɷтoн |šᵊtóm| < **ḫVtā́m*		[S] ɷтaн |šᵊtám| < **ḫVtḗm*	
t̲ms	[B] ɵɷнc |tʰómᵊs| < **t̲ám(V)s*	[B] ɵoнc |tʰóms(ᵊ)| < **t̲ámsVẏ*						
ḥms			[B] ꞓeнcı |hémsi| < **ḫÉms-Vt*		[F] ꞓнac |hᵊmás| < **ḫVmā́s*		[A] ꞓнec |hᵊmés| < **ḫVmḗs*	
šms				[S] ɷнɷe |šḿsə| < **šÉms-V(ẏ)-t*				
2⁺-EtR prtCpt	**Cá(XV)C / *CáC(VX)*	**CáwC-Vẏ*	**CÉ(X)C-Vt / *CÉC(X)-Vt*	**CÉ(X)C-ẏ-Vt / *CÉC(X)-ẏ-Vt*	**C¹VC²ā́C²*	**C¹VC²ā́C²-Vẏ*	**C¹VC²ḗC²*	**C¹VC²ḗC²-Vẏ*
šq	[S] ɷɷᴋ |šók| < **šá(XV)q / *šáq(VX)*		[S] ɷıкı |šíkʲ| < **šÉ(X)q-/*šÉq(X)-Vt*					
tt	[S] тɷт |tót| < **tá(XV)t / *tát(VX)*	[S] тooүтe |tówtə| < **táwt-Vẏ*						
d̲ẏ < *d̲d*	[S] ҳɷ |čó| < **d̲á(XV)ẏ / *d̲áẏ(VX)*	[A2] ҳooүe |čówə| < **d̲áwẏ-Vẏ*	[S] □ҳı |čí| < **d̲É(X)ẏ-/*d̲Éẏ(X)-Vt*					
pr		[A] пaүpe |páwrə| < **páwr-Vẏ*	[S] пıpe |pírə| < **pÉ(X)r-/ *pÉr(X)-Vt*	[S] пppe |pɾ́rə| < **pÉ(X)r-ẏ-Vt / *pÉr(X)-ẏ-Vt*				
ỉr			[S] eıpe |írə| < **iÉ(X)r-/*iÉr(X)-Vt*		[B] pa |rá| < **iVrā́r*			
ḫn	[S] ꞓɷn |hón| < **ḫá(XV)n / *ḫán(VX)*				[Sa] ꞓnaн |hᵊnán| < **ḫVnā́n*			
hr						[B] ꞓpoүp |hᵊrúr(ᵊ)| < **hVrā́r-Vẏ*		
t̲r							[S] тλλ |tᵊlél| < **t̲VrḖr*	[S] тeλнλ |təlél(ᵊ)| < **t̲VrḖr-Vẏ*

Table 7 | Full paradigm of infinitives/verbal nouns for 3-EtR^{prtCpt} and 2^{+}-EtR^{prtCpt} roots.

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